CIVIL ENGINEERING PORTFOLIO

QUANTITY SURVEYING, COST ESTIMATION & CONSTRUCTION MANAGEMENT

Comprehensive Compilation of Verified Engineering Deliverables, BNSP Level 5 State License, AutoCAD Multi-Viewport DED Drawings, Technical Specifications (RKS), BoQ/RAB Cost Estimations, S-Curve Scheduling, Bar Bending Schedules (BBS), and On-Site Quality Control.

REZA SEPDIDE, A.Md.T.

Civil Engineer | Quantity Surveyor | Construction Planner

3.72	LEVEL 5	10+	3 LANG.
GPA / 4.00	**SKK BNSP**	**PROJECTS**	**CERTIFIED**
D3 CIVIL ENG. PNP WITH HONORS	INTERMEDIATE BUILDING FIELD SUPERVISOR	REVITALIZATION, EPC QS, DED & S-CURVE	INDONESIAN, JAPANESE (SSW/JLPT), ENGLISH

CURRENT EDUCATION:	Diploma IV (D4) Construction Engineering Management, PNP (Active / Ongoing)
ACADEMIC DEGREE:	Diploma III (D3) Civil Engineering, Padang State Polytechnic (Graduated With Honors, GPA 3.72)
STATE LICENSE:	BNSP Level 5 Construction Services Center Region I Banda Aceh, Ministry of Public Works
EMAIL & CONTACT:	`rezasepdide@gmail.com` | `+62 896-4459-0915` | `linkedin.com/in/rezasepdide`

DOCUMENT INTEGRITY & DIGITAL VERIFICATION:

This portfolio presents verified engineering deliverables including authentic AutoCAD DED sheets, BoQ/RAB cost estimations conforming to PUPR AHSP standards, S-Curve & MS Project baselines, Bar Bending Schedules (BBS), and state licenses and international language certifications.

REZA SEPDIDE, A.Md.T.

Intermediate Building Field Supervisor

Reg. BNSP: F7472118 | KKNI Level 5

CORE ENGINEERING COMPETENCIES:

- Cost Estimation & BoQ/RAB PUPR AHSP Standards
- Bar Bending Schedule (BBS) Detailing
- Concrete Quality Control & Slump Testing
- S-Curve & Project Scheduling (CPM)
- AutoCAD Multi-Viewport DED Drawings to PDF
- Autodesk Revit BIM & SAP2000 Structural Modeling

EXECUTIVE PROFESSIONAL SUMMARY

Civil Engineering graduate from Padang State Polytechnic with honors (*With Honors*, GPA 3.72/4.00), currently pursuing a Bachelor of Applied Science (D4) in Construction Engineering Management. Certified with the Indonesian National Professional Certification Board (BNSP) Level 5 license as an Intermediate Building Construction Field Supervisor.

Demonstrates rigorous practical competence across Quantity Surveying (BoQ/RAB, AHSP), Bar Bending Schedule (BBS) detailing, S-Curve / MS Project schedule control, AutoCAD Multi-Viewport DED layouts to PDF, Autodesk Revit BIM 3D modeling, and SAP2000 structural analysis. Internationally certified bilingual in Japanese (SSW JAC, JFT-Basic 250, JLPT N4 165) and Professional English (TOEIC 640).

FORMAL EDUCATION TIMELINE

1. Diploma IV (D4) Construction Engineering Management — **Active / Ongoing**

Padang State Polytechnic (Aug 2026 — Present). Focus: Integrated Project Management, Cost/Quality Control, Construction Law, and Digital BIM.

2. Diploma III (D3) Civil Engineering — **With Honors (GPA 3.72)**

Padang State Polytechnic (2022 — 2025). Final Project: BoQ/RAB Cost Estimation & Time Scheduling for DPRD Pasaman Building.

3. SMAN 1 Padang (State Senior High School 1) — Natural Sciences (2019 — 2022)

CORE ENGINEERING COMPETENCIES & THESIS HIGHLIGHTS:

- **Precision Quantity Take-Off (QTO):** Mathematical precision in calculating earthworks, deep/shallow foundations, reinforced concrete superstructure, finishes, and MEP.
- **Schedule & Cost Control:** Translating WBS into Critical Path Method (CPM) networks and distributing weight percentages to cumulative S-Curves for variance control.
- **Specification Compliance:** Formulating RKS clauses, job safety protocols (SMKK), and site material testing checklists.

4 PILLARS OF ENGINEERING PRINCIPLES & ETHICS

01. STRUCTURAL INTEGRITY & QUALITY

Strict adherence to DED drawings and technical specifications. Fresh concrete slump testing (SNI 1972:2008), cylindrical compressive strength tests f'_c 25–30 MPa, concrete cover tolerances, and rebar lap splicing without compromise.

02. QUANTITY ACCURACY & BBS OPTIMIZATION

Bar Bending Schedule (BBS) optimization minimizing scrap waste ($< 3\%$). Producing transparent, auditable BoQ/RAB adhering to PUPR AHSP guidelines.

03. ADMINISTRATIVE COMPLIANCE & CCO RIGOR

Every progress billing is backed by verified backup volume calculations, accurate technical justifications for Contract Change Orders (CCO), and structured joint survey field logs.

04. GLOBAL STANDARDS & DEDICATION

Commitment to continuous development through D4 CEM studies, parametric Revit BIM adoption, SAP2000 structural modeling, and verified Japanese (SSW, JLPT) and English (TOEIC) certifications.

TECHNICAL SKILLS MATRIX, SOFTWARE & LANGUAGE QUALIFICATIONS

KKNI LEVEL 5 STANDARD | BIM, CAD, QS, QC & BILINGUAL

TECHNICAL ENGINEERING SKILLS

- **Quantity Take-Off (QTO) & BoQ:** Detailed volume calculations from DED sheets conforming to PUPR AHSP standards.
- **RKS Technical Specifications:** Drafting project work requirements, material quality criteria, and execution methods.
- **S-Curve Scheduling:** Work Breakdown Structure (WBS), weight distribution, CPM critical path, and EVM deviation control.
- **Bar Bending Schedule (BBS):** Rebar cutting optimization ensuring scrap waste $< 3\%$.
- **AutoCAD DED Drafting:** 2D architectural & structural drafting across multi-viewports scaled to PDF.
- **Quality Control (QC):** Fresh concrete slump tests, rebar placement checklists, and punch list administration.
- **Mutual Check & CCO:** MC-100 certification and Contract Change Order (CCO) variance management.

ENGINEERING SOFTWARE PROFICIENCY

- **AutoCAD (2D DED):** Advanced. Full 2D drafting, multi-viewport paper layout configuration, standard title blocks to PDF.
- **Microsoft Excel:** Advanced. Mathematical backup volume models, BoQ matrices, S-Curve scheduling algorithms.
- **Microsoft Project:** Intermediate. WBS hierarchy, CPM critical path, duration baselines, and tracking.
- **SAP2000:** Intermediate. 3D reinforced concrete space frames, dynamic response spectrum seismic loading (SNI).
- **Autodesk Revit (BIM):** Intermediate. 3D parametric building modeling of columns, beams, slabs, and architecture.
- **SketchUp:** Intermediate. 3D construction sequencing visualization for technical tender presentations.

CERTIFIED LANGUAGE CAPABILITIES

• **Indonesian:** Native fluency. Technical contractual correspondence, formal Minutes of Meeting (BA), and project dossiers.

• **Japanese:**

– **SSW Construction (JAC):** Passed official Japanese evaluation exam.

– **JFT-Basic:** Score 250 (Full Score, The Japan Foundation).

– **JLPT N4:** Score 165 (High Honors, The Japan Foundation).

• **English:**

– **TOEIC Score: 640 / 990:** Score 640 (Listening: 400, Reading: 240) Language Center, Padang State Polytechnic.

– Fluent comprehension of international engineering standards (ASTM, ACI, BS).

METHODOLOGY & QUALITY ASSURANCE:

• Field verification of elevations, structural member dimensions, rebar detailing, and concrete grade per SNI technical standards.

• Verification of concrete cover thickness, stirrup spacing, and double-strand tie-wire binding prior to concrete pouring approval.

DIGITALIZATION & BIM INTEGRATION:

• Direct integration of 3D parametric modeling and mathematical quantity takeoff to eliminate fabrication errors.

• Implementation of BIM Level 2 workflows for drawing coordination and inter-element clash detection.

MULTINATIONAL WORK READINESS:

• Fully prepared for deployment within Japanese contractor Joint Venture (JV) projects or multinational project teams.

• Fluent command of Japanese construction engineering terminology and Kiken Yochi (KYT) hazard prediction safety protocols.

BNSP LEVEL 5 STATE LICENSE

Intermediate Building Field Supervisor

Qualification: Technician / Analyst (Level 5)

Issuing Body:	BNSP RI
Implementing Agency:	BJKW I Banda Aceh (PUPR)
LSP Agency:	LSP Tenaga Konstruksi
Reg. Number:	F7472118
Validity Period:	Valid Thru July 2030 (5 Yrs)

TECHNICAL AUTHORITY SCOPE

Validates legal competence to direct and supervise multi-story building construction, oversee DED compliance, enforce SMKK safety, and conduct structural concrete QC.

LEGAL BASIS & STATE QUALIFICATION:

- Officially issued pursuant to Law No. 2/2017 regarding Construction Services.
- Serves as formal legal prerequisite for technical supervisors of multi-story building works in government and private sector tenders.

F7472118

BADAN NASIONAL SERTIFIKASI PROFESI
INDONESIA PROFESSIONAL CERTIFICATION AUTHORITY

SERTIFIKAT KOMPETENSI
CERTIFICATE OF COMPETENCE

Dengan ini menyatakan bahwa,
This is to certify that,

REZA SEPDIDE

Telah Kompeten pada bidang:
Is competent in the area of:

Jasa Konstruksi
Construction Services

Dengan Kualifikasi / Kompetensi:
With Qualification / Competency:

Pelaksana Lapangan Pekerjaan Gedung Madya
Implementing Intermediate Building Works

Sertifikat ini berlaku untuk 5 (lima) tahun
This certificate is valid for 5 (five) years

Atas nama Badan Nasional Sertifikasi Profesi
On Behalf of Indonesia Professional Certification Authority

Lembaga Sertifikasi Profesi Tenaga Konstruksi Nasional
Tenaga Konstruksi Nasional Professional Certification Agency

Bryan Matthew, ST
Ketua LSP
Chairman PCA

Official Document: SKK BNSP Level 5 Certificate of Competence — Front Page

SKKNI COMPETENCY UNIT STANDARDS

Assessed Competency Units

Intermediate Building Field Supervisor

Assessment Scheme:	Intermediate Building Field Supervisor
KKNI Level:	Level 5 (Technician / Analyst)
Assessment Status:	**All Competency Units Qualified (Certified)**
Testing Center (TUK):	BJKW I Banda Aceh

COMPETENCY UNIT SCOPE

Mastery of occupational safety and health (OSH/K3), earthworks/foundations, reinforced concrete structures (rebar, formwork, pouring), masonry/finishing, and project progress reporting administration.

BNSP ASSESSOR VERIFICATION ACCOUNTABILITY:

- All 10 competency units rigorously assessed via field practical observation, written tests, and portfolio interviews by BNSP-licensed assessors.
- Proves complete operational readiness to lead on-site field teams independently.

LEMBAGA PENGEMBANGAN JASA KONSTRUKSI
CONSTRUCTION SERVICES DEVELOPMENT BOARD

Daftar Unit Kompetensi:
List of Unit(s) of Competency:

Klasifikasi *Classification*	: *:*	Sipil *Civil*
Subklasifikasi *Subclassification*	: *:*	Gedung *Building*
Kualifikasi *Qualification*	: *:*	Teknisi/ Analis *Technician/ Analyst*
Jenjang *Level*	: *:*	5 (Lima) *5 (Five)*
Okupasi *Occupation*	: *:*	Pelaksana Lapangan Pekerjaan Gedung Madya *Implementing Intermediate Building Works*

Ditetapkan di Jakarta, 04 Juli 2025
Enacted in Jakarta, July 04, 2025

REZA SEPDIDE

Keterangan / *Remarks* :

1. Sertifikat ini sah berlaku setelah tercatat yang dibuktikan dengan nomor registrasi Sertifikat Kompetensi Kerja Konstruksi. / *This certificate is valid upon being registered as evidenced by registration number of Certificate of Competency of Contruction Works.*
2. QR Code dan Data yang tertera dalam sertifikat ini dapat diverifikasi melalui sistem informasi jasa konstruksi terintegrasi. / *QR Code and Data contained herein may be verified through an integrated information system of construction service.*

Official Document: SKK BNSP Level 5 Certificate of Competence — Units of Competency

REVIT BIM DIGITAL TRAINING

Revit Masterclass

Arsisten Academy (BIM 3D)

Academy:	Arsisten Academy
Scheme:	Revit Masterclass
Status:	**Completed with Competency**
Verification ID:	`eYfV8Gq3oA`

CURRICULUM & COMPETENCIES

Parametric 3D structural modeling (columns, beams, slabs), architectural elements, view templates, automated floor plans & elevations section cuts, and MTO material take-off schedules.

BIM LEVEL 2 FOUNDATION:

- Proficient in coordinated 3D parametric BIM modeling workflows.
- Capable of extracting automated material quantity schedules to eliminate volume discrepancies during bidding and execution.

Official Document: Certificate of Completion — Revit Masterclass for Beginners (Arsisten Academy)

JAPANESE CONSTRUCTION INDUSTRY STANDARDS

SSW Construction Industry

Civil Engineering (i) (Japanese)

Testing Body:	JAC (Japan)
Scope:	Civil Engineering (i)
Evaluation Result:	**Qualified / Passed**
Reg. Number:	40563425

QUALIFICATION SIGNIFICANCE

Official Specified Skilled Worker (*Tokutei Ginou*) evaluation exam supervised by MLIT Japan, validating technical competencies conforming to Japanese civil engineering standards.

JAPANESE CONSTRUCTION INDUSTRY STANDARDS:

- Deep understanding of Kiken Yochi Training (KYT) hazard prediction safety protocols and on-site 5S discipline.
- Fluent command of Japanese civil engineering terminology and complete readiness for deployment on Japanese-Indonesian Joint Venture projects.

建設分野特定技能評価試験結果通知書

Notification of results for Specified skills evaluation exam for Construction Industry

受験番号 Registration Number	: 40563425
氏名 Name	: REZA SEPDIDE ー
国籍 Nationality	: Indonesia
生年月日 Date of Birth	: 2003/09/28 (yyyy/mm/dd)
性別 Sex	: Male
受験地 Test Location	: Indonesia
受験日 Test Date	: 2026/07/11 (yyyy/mm/dd)

試験の結果は以下の通りです。

Your test results are as follows.

試験名 Exam Name	: Civil engineering (i) (Japanese)
試験結果 Exam Result	: Pass

【合格した方へ】
日本で特定技能外国人として働くには、この試験の合格に加えて、日本語試験にも合格しなければなりません。
建設分野では、在留資格申請の前に国土交通省への受入れ計画認定申請も必要です。
受入れ計画認定申請に使用する合格証明書が必要な方は、以下のページを参考にして、
建設技能人材機構のスマホアプリから合格証明書の発行申請を行ってください。

https://global.jac-skill.or.jp/news/manual/jac-members.php

【For those who passed the exam】
In order to work as a specified skills worker in Japan, in addition to passing this exam, you must also pass the Japanese language exam.
In the construction field, it is also necessary to apply for acceptance plan certification to the Ministry of Land, Infrastructure, Transport and Tourism before applying for a status of residence.
If you need a pass certificate to be used for acceptance plan certification application, please refer to the following page.
Please apply for issuance of a pass certificate from the smartphone app of the Japan Association for Construction Human Resources.

https://global.jac-skill.or.jp/news/manual/jac-members.php

Official Document: Specified Skilled Worker Evaluation Result — Construction Industry (JAC Japan)

JAPANESE LANGUAGE PROFICIENCY

The Japan Foundation

JFT-Basic Score 250 & JLPT N4 Score 165

1. JFT-BASIC (A2 STANDARD)

Score: **250 Points** (CEFR Level A2: 200 pass mark). Qualified in everyday conversation and on-site job instructions.

2. JLPT LEVEL N4

Score: **165 Points** (Passing Score: 90). Result: **Qualified** / **Passed (97.4th Percentile)**.

APLIKASI DWILINGUAL LAPANGAN:

- Daily operational communication with Japanese supervisors and engineers.
- Fully capable of understanding verbal/written instructions and interpreting Japanese technical work documentation and shop drawings.

JAPANFOUNDATION 国際交流基金

国際交流基金日本語基礎テスト 判定結果通知書

Japan Foundation Test for Basic Japanese Notification of assessment results

受験番号 Registration Number	: 40558178
氏名 Name	: REZA SEPDIDE —
国籍 Nationality	: Indonesia
生年月日 Date of Birth	: 2003/09/28 (yyyy/mm/dd)
性別 Sex	: Male
受験地 Test Location	: Indonesia
受験日 Test Date	: 2026/07/14 (yyyy/mm/dd)

試験の結果をお知らせします。
Your test results are as follows.

総合得点：250 点 Total Score points

(得点範囲：10 - 250 点, Range of Scores points 判定基準点： 200 点) Passing Score points

250
10 200 250

あなたは国際交流基金日本語基礎テストにおいて、ある程度日常会話ができ、生活に支障がない程度の日本語能力水準に達していると判定されました。
You were assessed to have reached a level of Japanese language proficiency in Japan Foundation Test for Basic Japanese to be able to engage in everyday conversation to a certain extent and without difficulties in daily life.

セクション毎の正答率は次のとおりです。
The percentage of correct answers for each section are as follows.

文字と語彙 Script and Vocabulary	100%
会話と表現 Conversation and Expression	100%
聴解 Listening Comprehension	100%
読解 Reading Comprehension	100%

Document 1: Official JFT-Basic Test Result (Score 250)

N4

日本語能力認定書

CERTIFICATE
JAPANESE-LANGUAGE PROFICIENCY

氏名 Name	REZA SEPDIDE
生年月日 (y/m/d) Date of Birth	2003/09/28
受験地 Test Site	インドネシア Indonesia

上記の者は2026年7月に独立行政法人国際交流基金および
公益財団法人日本国際教育支援協会が実施した日本語能力試験
N4 レベルに合格したことを証明します。

2026年8月18日

This is to certify that the person named above has passed Level N4 of the Japanese-Language Proficiency Test given in July 2026, jointly administered by the Japan Foundation and Japan Educational Exchanges and Services.

August 18, 2026

独立行政法人 国際交流基金 The Japan Foundation

公益財団法人 日本国際教育支援協会 Japan Educational Exchanges and Services

N4A499511A

26A2010601-40410

Document 2: Official JLPT N4 Certificate (Score 165)

INDUSTRIAL FIELD INTERNSHIP

PT Solusi Industri Energi

Waste Gas Recovery Project (WGR)

Contractor:	PT Solusi Industri Energi
Client:	PT Indo-Bharat Rayon
Location:	Purwakarta, West Java
Penilaian:	**Sangat Baik (A)**

CORE FIELD CONTRIBUTIONS

Bar Bending Schedule (BBS) for heavy machine foundations, rework BoQ estimation, fresh concrete slump tests, and daily, weekly, and monthly progress reporting.

PENGAKUAN KINERJA INDUSTRI:

- Awarded Grade A (Outstanding) for dedication and technical precision in rebar fabrication and site supervision.
- Proven operational ability within high-risk industrial plant environments.

Official Document: Industrial Internship Certificate — PT Solusi Industri Energi

TECHNICAL ENGLISH PROFICIENCY

TOEIC Score: 640 / 990

Total Score: 640 Points

Testing Center:	Language Center, PNP Padang
Listening:	**400 Poin**
Reading:	**240 Poin**
Total Score:	**640 Poin**

ENGINEERING PROFESSIONAL APPLICATION

Interpreting technical manuals, international literature, operating engineering software (Revit, SAP2000), and conducting international engineering correspondence.

TECHNICAL ENGLISH MASTERY:

- Equivalent to CEFR B1/B2 Working Proficiency for interpreting international construction codes (ASTM, ACI, BS).
- Effective professional communication within multinational project management teams.

POLITEKNIK NEGERI PADANG
UPT BAHASA

NOTIFICATION LETTER

PREDICTION TEST SCORE FOR TOEIC® TEST

No. 002/PL9.13/KM/2023

This is to certify that

REZA SEPDIDE
(Reg Number: 2201022034)

followed Prediction Test for the TOEIC Test ® on Mar 19, 2023 at UPT Bahasa Politeknik Negeri Padang and achieved the following score:

Listening	400
Reading	240
Total Score	640

Padang, Mar 20, 2023

SUMIRA, M.Pd
NIP. 197804012009122003

*For Internal use only

TOEIC® is registered trademark of Educational Testing Service (ETS)
This program is not endorsed or approved by ETS

Official Document: Official Notification Letter of TOEIC Prediction Test Score (Language Center, PNP)

ACADEMIC EXCELLENCE & HONORS

D3 Civil Engineering PNP

GPA: 3.72 / 4.00 (Graduated With Honors)

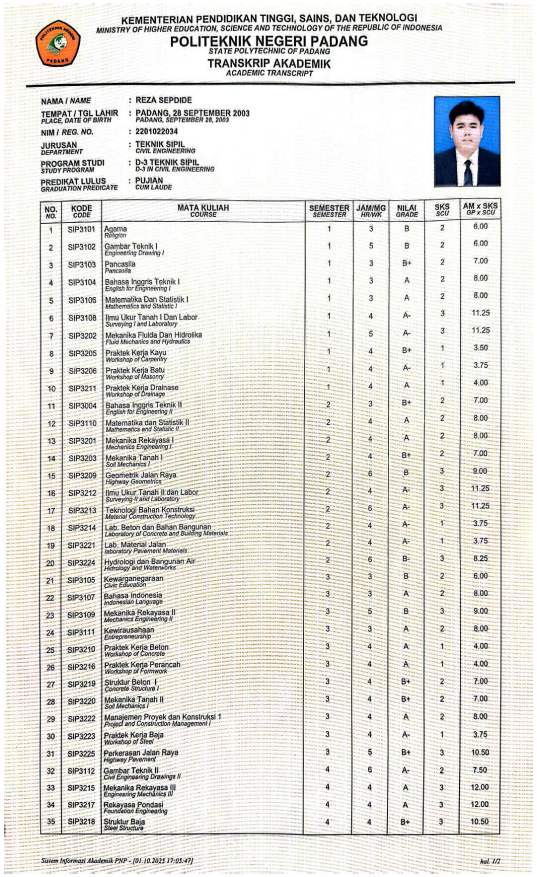
Institution:	Padang State Polytechnic
Cumulative GPA:	**3.72 / 4.00**
Distinction:	**With Honors (Cum Laude)**
Credit Load:	112 Credits Completed

KEY ENGINEERING COURSEWORK

Reinforced Concrete Structures, Cost Estimation & BoQ, Construction Management & S-Curve, Engineering CAD Drafting, Soil Mechanics, Foundation Design, and Structural Analysis.

CORE ENGINEERING COURSE PERFORMANCE:

- Achieved Grade A in core courses: Reinforced Concrete Design, Cost Estimation & BoQ, Construction Management, and Structural Modeling.
- Demonstrates robust theoretical mastery and rigorous technical calculation competencies.

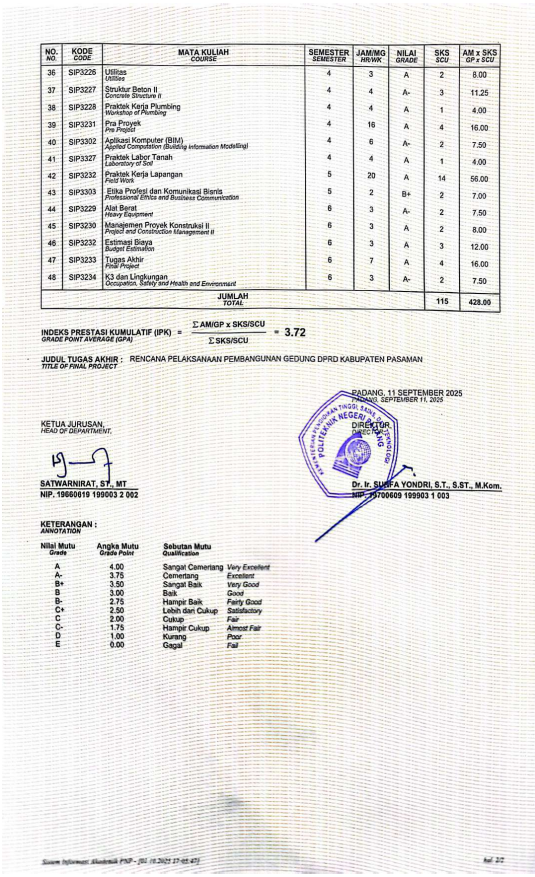


KEMENTERIAN PENDIDIKAN TINGGI, SAINS, DAN TEKNOLOGI
MINISTRY OF HIGHER EDUCATION, SCIENCE AND TECHNOLOGY OF THE REPUBLIC OF INDONESIA
POLITEKNIK NEGERI PADANG
STATE POLYTECHNIC OF PADANG
TRANSKRIP AKADEMIK
ACADEMIC TRANSCRIPT

NAMA / NAME : REZA SEPDIDE
TEMPAT / TGL LAHIR : PADANG, 28 SEPTEMBER 2003
PLACE, DATE OF BIRTH : PADANG, SEPTEMBER 28, 2003
NIM / REG. NO. : 2201022034
JURUSAN : TEKNIK SIPIL
DEPARTMENT : CIVIL ENGINEERING
PROGRAM STUDI : D-3 TEKNIK SIPIL
STUDY PROGRAM : D-3 IN CIVIL ENGINEERING
PREDIKAT LULUS : PUJIAN
GRADUATION PREDICATE : CUM LAUDE

NO.	KODE	MATA KULIAH	SEMESTER	JAM/MG	NILAI	SKS	AM x SKS
NO.	CODE	COURSE	SEMESTER	HR/WK	GRADE	SCU	GP x SCU
1	SIP3101	Agama Religion	1	3	B	2	6.00
2	SIP3102	Gambar Teknik I Engineering Drawing I	1	5	B	2	6.00
3	SIP3103	Pancasila Pancasila	1	3	B+	2	7.00
4	SIP3104	Bahasa Inggris Teknik I English for Engineering I	1	3	A	2	8.00
5	SIP3106	Matematika Dan Statistik I Mathematics and Statistic I	1	3	A	2	8.00
6	SIP3108	Ilmu Ukur Tanah I Dan Labor Surveying I and Laboratory	1	4	A-	3	11.25
7	SIP3202	Mekanika Fluida Dan Hidrolika Fluid Mechanics and Hydraulics	1	5	A-	3	11.25
8	SIP3205	Praktek Kerja Kayu Workshop of Carpentry	1	4	B+	1	3.50
9	SIP3206	Praktek Kerja Batu Workshop of Masonry	1	4	A-	1	3.75
10	SIP3211	Praktek Kerja Drainase Workshop of Drainage	1	4	A	1	4.00
11	SIP3004	Bahasa Inggris Teknik II English for Engineering II	2	3	B+	2	7.00
12	SIP3110	Matematika dan Statistik II Mathematics and Statistic II	2	4	A	2	8.00
13	SIP3201	Mekanika Rekayasa I Mechanics Engineering I	2	4	A-	2	8.00
14	SIP3203	Mekanika Tanah I Soil Mechanics I	2	4	B+	2	7.00
15	SIP3209	Geometrik Jalan Raya Highway Geometrics	2	6	B	3	9.00
16	SIP3212	Ilmu Ukur Tanah II dan Labor Surveying II and Laboratory	2	4	A-	3	11.25
17	SIP3213	Teknologi Bahan Konstruksi Material Construction Technology	2	6	A-	3	11.25
18	SIP3214	Lab. Beton dan Bahan Bangunan Laboratory of Concrete and Building Materials	2	4	A-	1	3.75
19	SIP3221	Lab. Material Jalan laboratory Pavement Materials	2	4	A-	1	3.75
20	SIP3224	Hydrologi dan Bangunan Air Hidrology and Waterworks	2	6	B-	3	8.25
21	SIP3105	Kewarganegaraan Civic Education	3	3	B	2	6.00
22	SIP3107	Bahasa Indonesia Indonesian Language	3	3	A	2	8.00
23	SIP3109	Mekanika Rekayasa II Mechanics Engineering II	3	5	B	3	9.00
24	SIP3111	Kewirausahaan Entrepreneurship	3	3	A	2	8.00
25	SIP3210	Praktek Kerja Beton Workshop of Concrete	3	4	A	1	4.00
26	SIP3216	Praktek Kerja Perancah Workshop of Formwork	3	4	A	1	4.00
27	SIP3219	Struktur Beton I Concrete Structure I	3	4	B+	2	7.00
28	SIP3220	Mekanika Tanah II Soil Mechanics I	3	4	B+	2	7.00
29	SIP3222	Manajemen Proyek dan Konstruksi 1 Project and Construction Management I	3	4	A	2	8.00
30	SIP3223	Praktek Kerja Baja Workshop of Steel	3	4	A-	1	3.75
31	SIP3225	Perkerasan Jalan Raya Highway Pavement	3	5	B+	3	10.50
32	SIP3112	Gambar Teknik II Civil Engineering Drawings II	4	6	A-	2	7.50
33	SIP3215	Mekanika Rekayasa III Engineering Mechanics III	4	4	A	3	12.00
34	SIP3217	Rekayasa Pondasi Foundation Engineering	4	4	A	3	12.00
35	SIP3218	Struktur Baja Steel Structure	4	4	B+	3	10.50

Official Academic Transcript — Page 1



NO.	KODE	MATA KULIAH	SEMESTER	JAM/MG	NILAI	SKS	AM x SKS
NO.	CODE	COURSE	SEMESTER	HR/WK	GRADE	SCU	GP x SCU
36	SIP3226	Utilitas Utilities	4	3	A	2	8.00
37	SIP3227	Struktur Beton II Concrete Structure II	4	4	A-	3	11.25
38	SIP3228	Praktek Kerja Plumbing Workshop of Plumbing	4	4	A	1	4.00
39	SIP3231	Pra Proyek Pre Project	4	16	A	4	16.00
40	SIP3302	Aplikasi Komputer (BIM) Applied Computation (Building Information Modeling)	4	6	A-	2	7.50
41	SIP3327	Praktek Labor Tanah Laboratory of Soil	4	4	A	1	4.00
42	SIP3232	Praktek Kerja Lapangan Field Work	5	20	A	14	56.00
43	SIP3303	Etika Profesi dan Komunikasi Bisnis Professional Ethics and Business Communication	5	2	B+	2	7.00
44	SIP3229	Alat Berat Heavy Equipment	6	3	A-	2	7.50
45	SIP3230	Manajemen Proyek Konstruksi II Project and Construction Management II	6	3	A	2	8.00
46	SIP3232	Estimasi Biaya Budget Estimation	6	3	A	3	12.00
47	SIP3233	Tugas Akhir Final Project	6	7	A	4	16.00
48	SIP3234	K3 dan Lingkungan Occupation, Safety and Health and Environment	6	3	A-	2	7.50
		JUMLAH TOTAL				115	428.00

INDEKS PRESTASI KUMULATIF (IPK) = $\frac{\Sigma \text{ AM/GP x SKS/SCU}}{\Sigma \text{ SKS/SCU}} = 3.72$
GRADE POINT AVERAGE (GPA)

JUDUL TUGAS AKHIR : RENCANA PELAKSANAAN PEMBANGUNAN GEDUNG DPRD KABUPATEN PASAMAN
TITLE OF FINAL PROJECT

PADANG, 11 SEPTEMBER 2025
PADANG, SEPTEMBER 11, 2025

KETUA JURUSAN,
HEAD OF DEPARTMENT,

SATWARNIRAT, ST., MT
NIP. 19660619 199003 2 002

DIREKTUR,
DIRECTOR,

Dr. Ir. SUHFA YONDRI, S.T., S.ST., M.Kom.
NIP. 19700609 199903 1 003

KETERANGAN :
ANNOTATION

Nilai Mutu Grade	Angka Mutu Grade Point	Sebutan Mutu Qualification	
A	4.00	Sangat Cemerlang	*Very Excellent*
A-	3.75	Cemerlang	*Excellent*
B+	3.50	Sangat Baik	*Very Good*
B	3.00	Baik	*Good*
B-	2.75	Hampir Baik	*Fairly Good*
C+	2.50	Lebih dari Cukup	*Satisfactory*
C	2.00	Cukup	*Fair*
C-	1.75	Hampir Cukup	*Almost Fair*
D	1.00	Kurang	*Poor*
E	0.00	Gagal	*Fail*

Official Academic Transcript — Page 2

TECHNICAL PLANNING PROJECT 01

SDN 19 Kuamang Revitalization

Engineering Consultant / Lead Planner

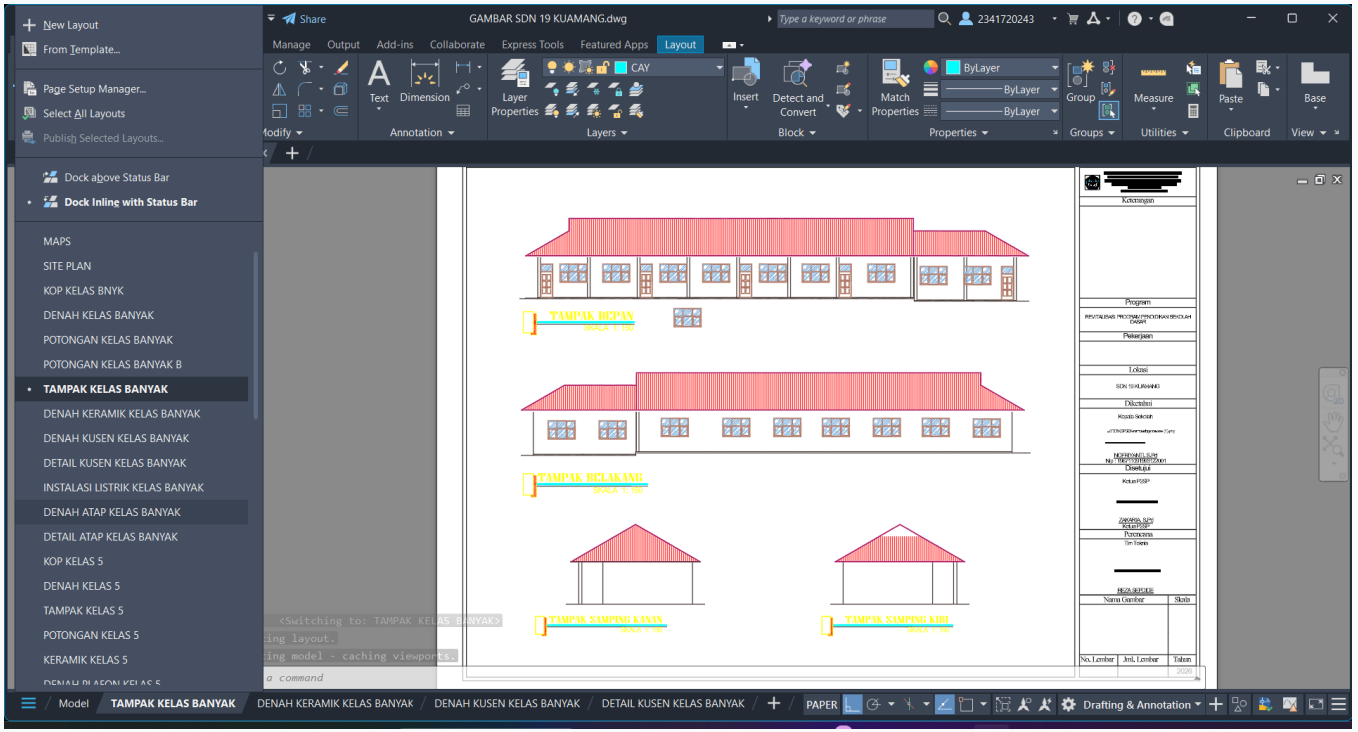
Role:	Lead Technical Planner
Location:	Pasaman, West Sumatra
Timeline:	September 2026
Deliverables:	DED, BoQ/RAB, RKS, S-Curve

AUTOCAD PAPER SPACE LAYOUT ARCHITECTURE

Multi-scale viewport configuration (1:100, 1:20), standard PUPR title blocks, standardized plot styles (CTB), and high-precision direct PDF printing.

MULTI-VIEWPORT DED PROTOCOLS:

- Strict separation of 1:1 Model Space and scaled Paper Space layout ensures absolute title block consistency.
- Guarantees sharp dimension notations and vector print resolution ready for field construction execution.



Working Drawing Evidence: AutoCAD Paper Space Layout Configuration for DED PDF Publishing (SDN 19 Kuamang)

ARCHITECTURAL & STRUCTURAL DED SHEETS

AutoCAD DED Sheets

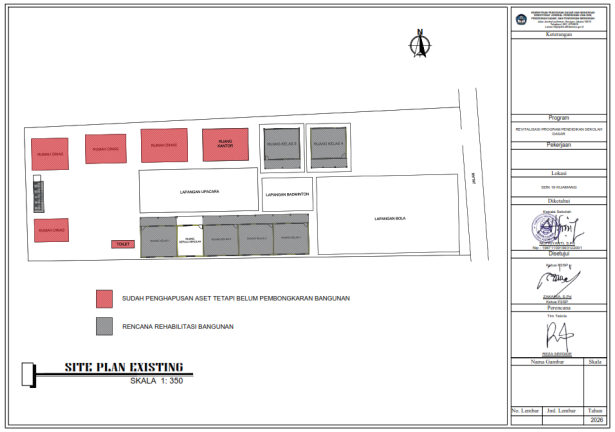
Classroom Rehabilitation DED

Detailed engineering design drawings serving as physical construction references for contractors:

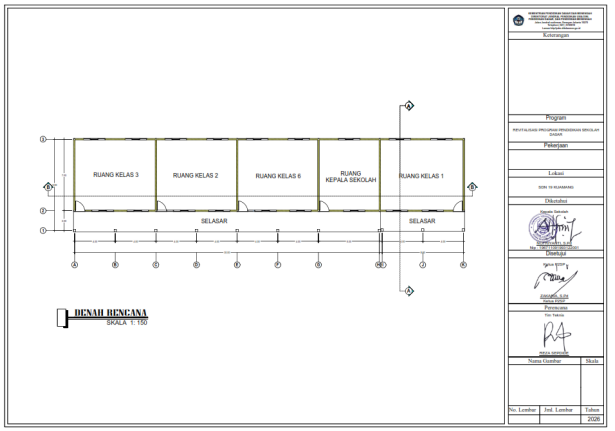
- **1. Floor Plan:** Spatial layout of classrooms, teacher offices, and natural ventilation corridors.
- **2. Architectural Elevations:** Front and side building elevations.
- **3. Roof Framing Plan:** Class-II timber truss load-bearing framework.
- **4. Ceiling Details:** Galvanized hollow framing pattern and sanitation plumbing.

REHABILITATION TECHNICAL SPECIFICATIONS:

- Replacement of structural Class-II timber trusses, new gypsum board ceiling on galvanized hollow steel framework.
- Practical column structural stiffening, ceramic floor rejuvenation, and full sanitation facility upgrades.

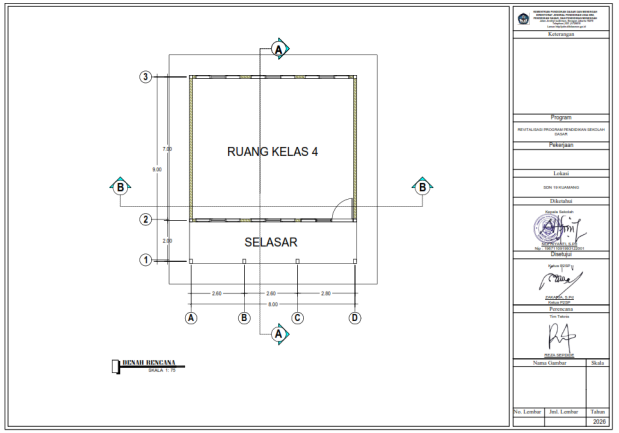


1. Architectural Floor Plan & Spatial Layout



2. Front & Side Building Elevations

3. Roof Timber Truss & Framing Layout



4. Ceiling Framing & Sanitation Details

TECHNICAL SPECIFICATIONS & QS

RKS, Volume & S-Curve

SDN 19 Kuamang, Pasaman

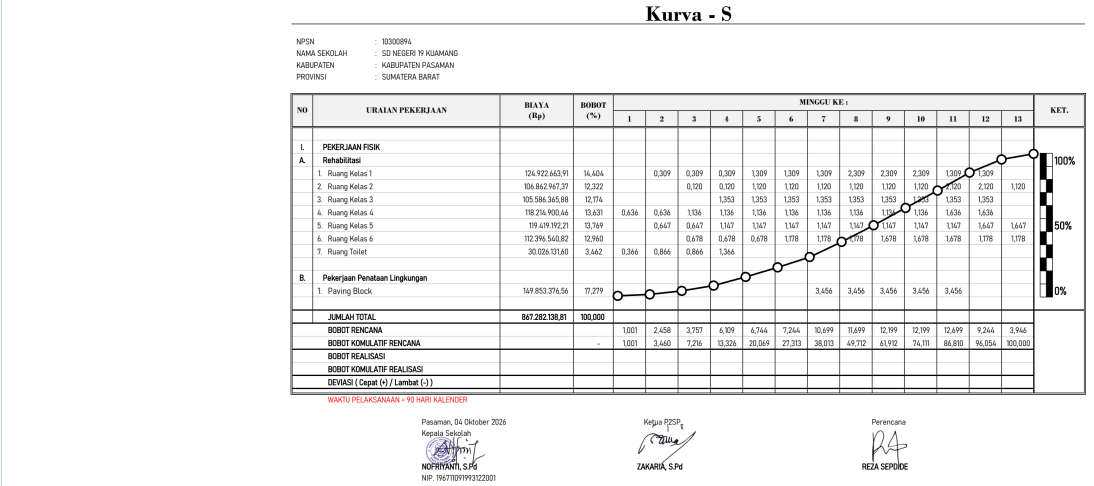
Scheme:	School Revitalization 2026
Total Budget:	Rp 967,660,000,00
Physical Cost:	Rp 867,282,138,81
Planned Time:	90 Days (13 Weeks)

QUALITY CONTROL INSTRUMENTS

- **RKS Specs:** Establishing SNI mortar mixes, Class II timber trusses, and site testing standards.
- **Backup Volume:** Mathematical calculation per room, toilet, and paving block as basis for BoQ.
- **S-Curve:** Cumulative physical progress mapping for early variance detection.

BUDGET & SCHEDULE INTEGRITY:

- **Cross-Validation of DED & BoQ:** Synchronizing backup volume takeoff directly with DED sheets to eliminate quantity variances.
- **Realistic Weight Distribution:** Establishing logical dependencies across structural and architectural activities to ensure adherence to the 90-calendar-day deadline.
- **Official Validation:** Formally endorsed and signed by the School Revitalization Committee (P2SP), School Principal, and Reza Sepdide (Lead Technical Planner).



Kurva - S

NPSN : 10300894
NAMA SEKOLAH : SD NEGERI 19 KUAMANG
KABUPATEN : KABUPATEN PASAMAN
PROVINSI : SUMATERA BARAT

NO	URAIAN PEKERJAAN	BIAYA (Rp)	BOBOT (%)	MINGGU KE : 1	2	3	4	5	6	7	8	9	10	11	12	13	KET.
I.	PEKERJAAN FISIK																
A.	Rehabilitasi																
1.	Ruang Kelas 1	124.922.663,91	14,404		0,309	0,309	0,309	1,309	1,309	1,309	2,309	2,309	2,309	1,309	1,309		
2.	Ruang Kelas 2	106.862.967,37	12,322			0,120	0,120	1,120	1,120	1,120	1,120	1,120	1,120	2,120	2,120	1,120	
3.	Ruang Kelas 3	105.586.365,88	12,174				1,353	1,353	1,353	1,353	1,353	1,353	1,353	1,353	1,353		
4.	Ruang Kelas 4	118.214.900,46	13,631	0,636	0,636	1,136	1,136	1,136	1,136	1,136	1,136	1,136	1,136	1,636	1,636		
5.	Ruang Kelas 5	119.419.192,21	13,769		0,647	0,647	1,147	1,147	1,147	1,147	1,147	1,147	1,147	1,147	1,647	1,647	
6.	Ruang Kelas 6	112.396.540,82	12,960			0,678	0,678	0,678	1,178	1,178	1,178	1,678	1,678	1,678	1,178	1,178	
7.	Ruang Toilet	30.026.131,60	3,462	0,366	0,866	0,866	1,366										
B.	Pekerjaan Penataan Lingkungan																
1.	Paving Block	149.853.376,56	17,279							3,456	3,456	3,456	3,456	3,456			
	JUMLAH TOTAL	867.282.138,81	100,000														
	BOBOT RENCANA			1,001	2,458	3,757	6,109	6,744	7,244	10,699	11,699	12,199	12,199	12,699	9,244	3,946	
	BOBOT KOMULATIF RENCANA		-	1,001	3,460	7,216	13,326	20,069	27,313	38,013	49,712	61,912	74,111	86,810	96,054	100,000	
	BOBOT REALISASI																
	BOBOT KOMULATIF REALISASI																
	DEVIASI (Cepat (+) / Lambat (-))																

WAKTU PELAKSANAAN = 90 HARI KALENDER

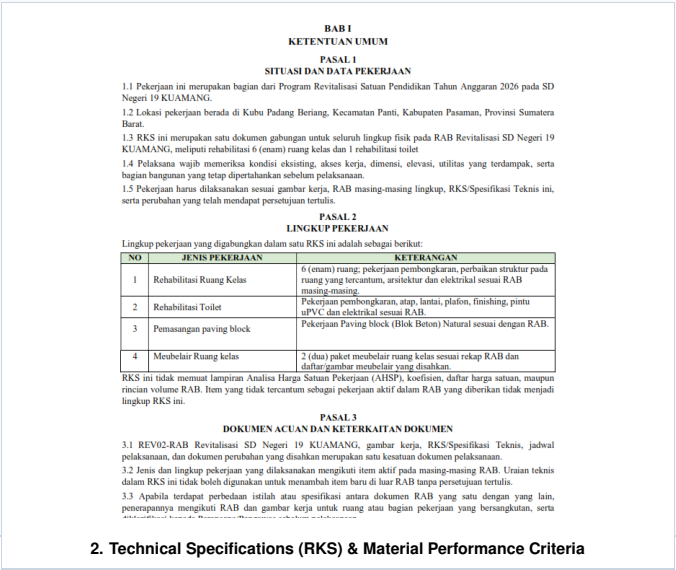
100%
50%
0%

Pasaman, 04 Oktober 2026
Kepala Sekolah
NOFRIYANTI, S.Pd
NIP. 196711091993122001

Ketua P2SP
ZAKARIA, S.Pd

Perencana
REZA SEPDIDE

1. S-Curve Progress Baseline & Weekly Weight Distribution Schedule (Total: Rp 867,282,138.81 — Duration: 13 Weeks)



BAB I
KETENTUAN UMUM

PASAL 1
SITUASI DAN DATA PEKERJAAN

1.1 Pekerjaan ini merupakan bagian dari Program Revitalisasi Satuan Pendidikan Tahun Anggaran 2026 pada SD Negeri 19 KUAMANG.

1.2 Lokasi pekerjaan berada di Kubu Padang Beriang, Kecamatan Panti, Kabupaten Pasaman, Provinsi Sumatera Barat.

1.3 RKS ini merupakan satu dokumen gabungan untuk seluruh lingkup fisik pada RAB Revitalisasi SD Negeri 19 KUAMANG, meliputi rehabilitasi 6 (enam) ruang kelas dan 1 rehabilitasi toilet

1.4 Pelaksana wajib memeriksa kondisi eksisting, akses kerja, dimensi, elevasi, utilitas yang terdampak, serta bagian bangunan yang tetap dipertahankan sebelum pelaksanaan.

1.5 Pekerjaan harus dilaksanakan sesuai gambar kerja, RAB masing-masing lingkup, RKS/Spesifikasi Teknis ini, serta perubahan yang telah mendapat persetujuan tertulis.

PASAL 2
LINGKUP PEKERJAAN

Lingkup pekerjaan yang digabungkan dalam satu RKS ini adalah sebagai berikut:

NO	JENIS PEKERJAAN	KETERANGAN
1	Rehabilitasi Ruang Kelas	6 (enam) ruang; pekerjaan pembongkaran, perbaikan struktur pada ruang yang tercantum, arsitektur dan elektrikal sesuai RAB masing-masing.
2	Rehabilitasi Toilet	Pekerjaan pembongkaran, atap, lantai, plafon, finishing, pintu uPVC dan elektrikal sesuai RAB.
3	Pemasangan paving block	Pekerjaan Paving block (Blok Beton) Natural sesuai dengan RAB.
4	Meubelair Ruang kelas	2 (dua) paket meubelair ruang kelas sesuai rekap RAB dan daftar/gambar meubelair yang disahkan.

RKS ini tidak memuat lampiran Analisa Harga Satuan Pekerjaan (AHSP), koefisien, daftar harga satuan, maupun rincian volume RAB. Item yang tidak tercantum sebagai pekerjaan aktif dalam RAB yang diberikan tidak menjadi lingkup RKS ini.

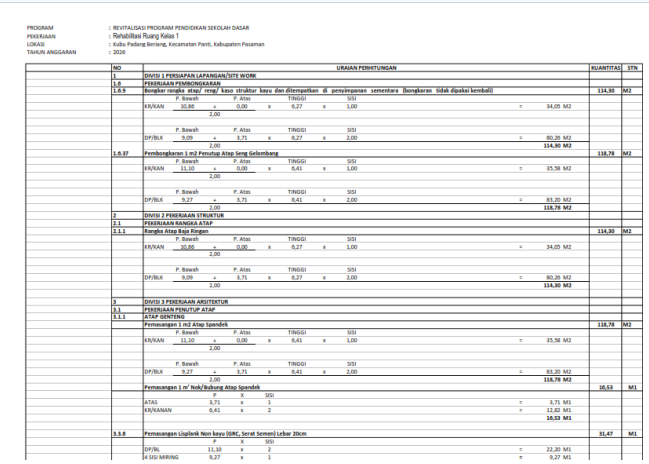
PASAL 3
DOKUMEN ACUAN DAN KETERKAITAN DOKUMEN

3.1 REV02-RAB Revitalisasi SD Negeri 19 KUAMANG, gambar kerja, RKS/Spesifikasi Teknis, jadwal pelaksanaan, dan dokumen perubahan yang disahkan merupakan satu kesatuan dokumen pelaksanaan.

3.2 Jenis dan lingkup pekerjaan yang dilaksanakan mengikuti item aktif pada masing-masing RAB. Uraian teknis dalam RKS ini tidak boleh digunakan untuk menambah item baru di luar RAB tanpa persetujuan tertulis.

3.3 Apabila terdapat perbedaan istilah atau spesifikasi antara dokumen RAB yang satu dengan yang lain, penerapannya mengikuti RAB dan gambar kerja untuk ruang atau bagian pekerjaan yang bersangkutan, serta

2. Technical Specifications (RKS) & Material Performance Criteria



PROGRAM : REVITALISASI PROGRAM PENDIDIKAN SEKOLAH DASAR
PEKERJAAN : Rehabilitasi Ruang Kelas 1
LOKASI : Kubu Padang Beriang, Kecamatan Panti, Kabupaten Pasaman
TAHUN ANGGARAN : 2026

NO	URAIAN PERHITUNGAN	KUANTITAS	STN
I.	DIVISI 1 PERSIAPAN LAPANGAN/SITE WORK		
I.0	PEKERJAAN PEMBONGKARAN		
I.0.9	Bongkar rangka atap/ ring/ kaso struktur kayu dan ditempatkan di penyimpanan sementara (bongkaran tidak dipakai kembali)	114,30	M2
	P. Bawah P. Atas TINGGI SISI KR/KAN 10,80 + 0,00 x 6,27 x 1,00 = 34,05 M2 2,00		
	P. Bawah P. Atas TINGGI SISI DP/BLK 5,09 + 5,71 x 6,27 x 2,00 = 80,26 M2 2,00		
	114,30 M2		
I.0.37	Pembongkaran 1 m2 Penutup Atap Seng Gelombang	118,78	M2
	P. Bawah P. Atas TINGGI SISI KR/KAN 11,10 + 0,00 x 6,41 x 1,00 = 35,58 M2 2,00		
	P. Bawah P. Atas TINGGI SISI DP/BLK 9,27 + 5,71 x 6,41 x 2,00 = 83,20 M2 2,00		
	118,78 M2		
2	DIVISI 2 PEKERJAAN STRUKTUR		
2.1	PEKERJAAN RANGKA ATAP		
2.1.1	Rangka Atap Baja Ringan	114,30	M2
	P. Bawah P. Atas TINGGI SISI KR/KAN 10,80 + 0,00 x 6,27 x 1,00 = 34,05 M2 2,00		
	P. Bawah P. Atas TINGGI SISI DP/BLK 5,09 + 5,71 x 6,27 x 2,00 = 80,26 M2 2,00		
	114,30 M2		
3	DIVISI 3 PEKERJAAN ARSITEKTUR		
3.1	PEKERJAAN PENUTUP ATAP		
3.1.1	ATAP GENTING		
	Pemasangan 1 m2 Atap Spandek	118,78	M2
	P. Bawah P. Atas TINGGI SISI KR/KAN 11,10 + 0,00 x 6,41 x 1,00 = 35,58 M2 2,00		
	P. Bawah P. Atas TINGGI SISI DP/BLK 9,27 + 5,71 x 6,41 x 2,00 = 83,20 M2 2,00		
	118,78 M2		
	Pemasangan 1 m' Nok/Bubung Atap Spandek	10,53	M1
	P X SISI ATAS 3,71 x 1 = 3,71 M1 KR/KANAN 6,41 x 2 = 12,82 M1		
	10,53 M1		
3.3.8	Pemasangan Lisplank Non kayu (GRC, Serat Semen) Lebar 20cm	31,47	M1
	P X SISI DP/BL 11,10 x 2 = 22,20 M1 & SISI MIRING 9,27 x 1 = 9,27 M1		

3. Detailed Backup Volume Quantity Take-Off Sheet for BoQ

TECHNICAL PLANNING PROJECT 02

SLBN 2 Pariaman Revitalization

Inclusive Special Education Facility

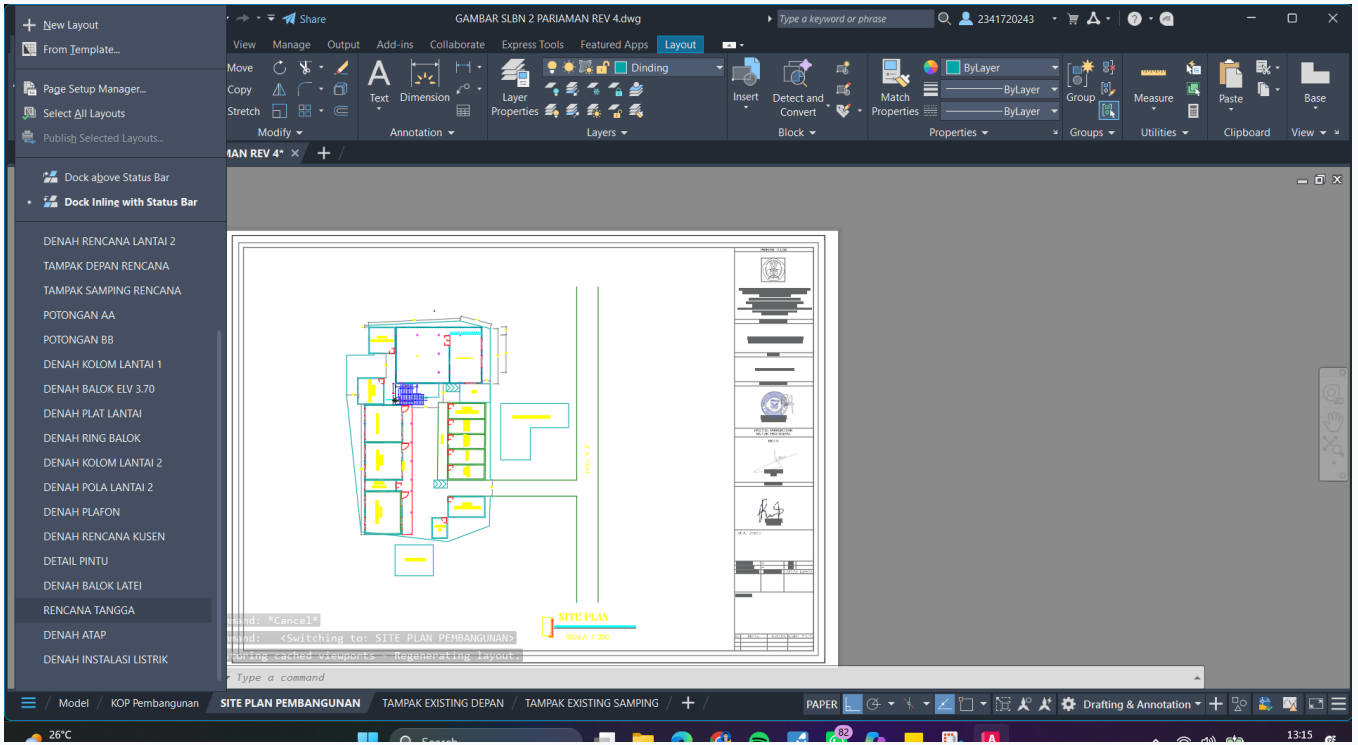
Role:	Lead Technical Planner
Location:	Pariaman City, West Sumatra
Facility:	Disability-Accessible Learning Unit
Deliverables:	DED, CCO Addendum, Volume, BoQ

INCLUSIVE DESIGN CHALLENGES

Safe-gradient wheelchair accessibility ramps ($< 1 : 12$), widened door clearances, dedicated disability-accessible sanitation, and standardized AutoCAD paper space layouts.

DISABILITY ACCESSIBILITY DESIGN CODES:

- Full compliance with PUPR Ministerial Regulation No. 14/PRT/M/2017 on Universal Building Accessibility.
- Slip-resistant wheelchair ramp textures, continuous double handrails, and sliding accessible restroom doors.



Working Drawing Evidence: AutoCAD Multi-Viewport Paper Space Layout (SLBN 2 Pariaman)

ARCHITECTURAL & STRUCTURAL DED DETAILS

Detailed Engineering Drawings (DED)

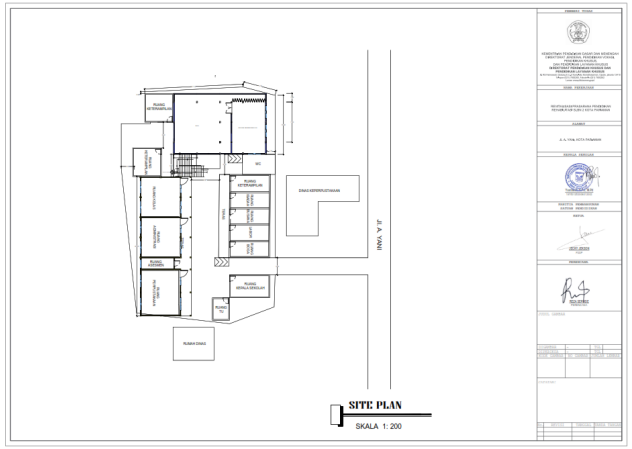
SLBN 2 Pariaman

Detailed engineering working drawings for on-site construction execution:

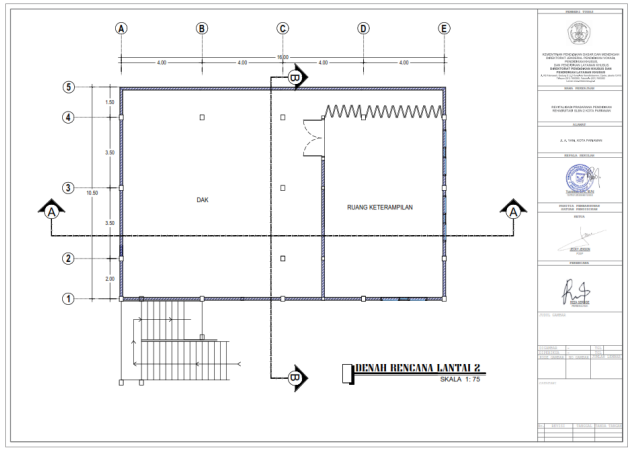
- **1. Floor Plan & Zoning:** Existing layout and renovated inclusive learning units.
- **2. Architectural Elevations:** Front and side facade elevations.
- **3. Ceiling Framing:** Galvanized hollow layout and acoustic board ceiling.
- **4. Structural Details:** Practical tie columns, lintel beams, and accessible plumbing.

STRUCTURAL REINFORCEMENT DETAILS:

- Rebar detailing for lintel beams above widened accessible door openings.
- Practical tie columns for masonry wall seismic reinforcement detailed at clear 1:20 enlarged scale.

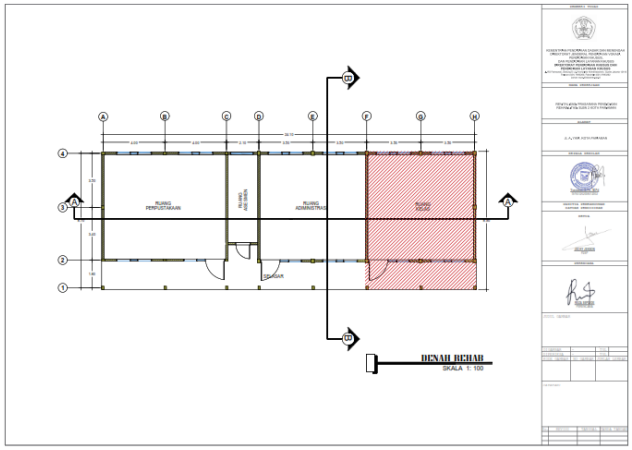


1. Existing Floor Plan & Classroom Zoning Plan



2. Architectural Facade Elevations

3. Ceiling Hollow Framing & Panel Layout



4. Accessible Sanitation & Structural Practical Details

STRUCTURAL DOCUMENTATION & CCO 02

CCO & Backup Volume

SLBN 2 Pariaman, West Sumatra

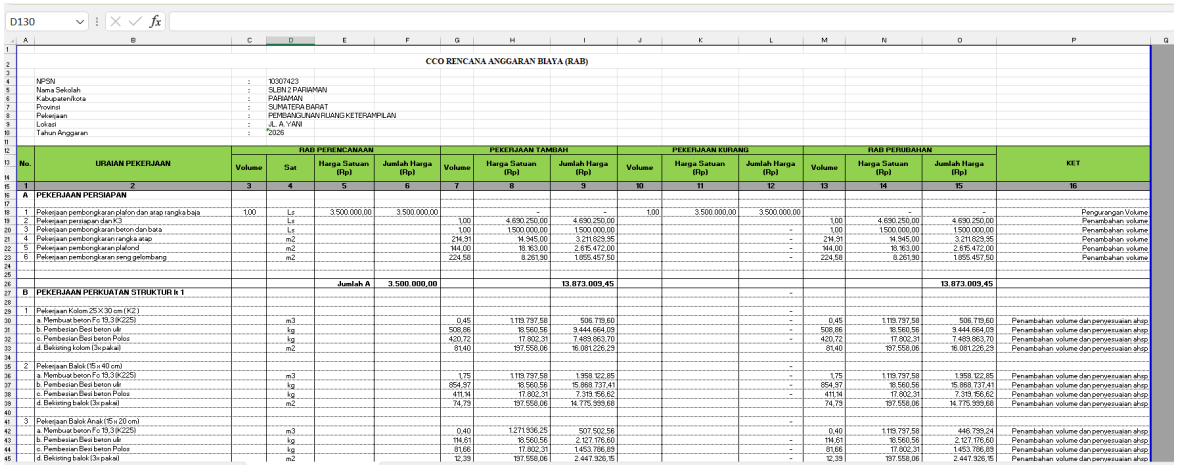
Scheme:	School Revitalization Grant 2026
Document:	CCO Addendum & BoQ
Adjustment:	Site Ground Elevation
Total BoQ:	Rp 1,046,960,116,8

FIELD JUSTIFICATION

- **CCO Addendum:** Re-balancing quantity additions and deductions due to natural site ground contours.
- **Revised Elevation:** Modifying wall heights, practical columns, and wheelchair ramp access.
- **Backup Volume:** Mathematical calculation for plaster, screed, tiling, and painting.

CCO ACCOUNTABILITY:

- **Contractual Compliance:** Addendum documentation structured strictly per Indonesian public works construction contract regulations.
- **Budget Neutrality:** Reconciled quantity additions and deductions balanced by finishing efficiency to maintain total allocated budget ceiling.
- **Formal Endorsement:** Legally signed and certified by the School Principal, Construction Committee (P2S), and Reza Sepdide (Lead Technical Planner).

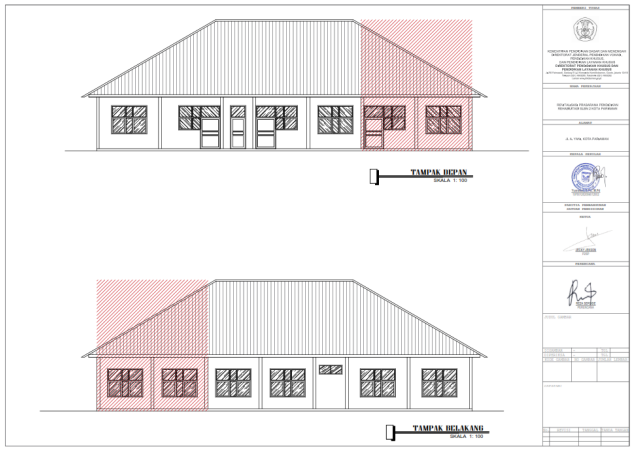


CCO RENCANA ANGGARAN BIAYA (RAB)

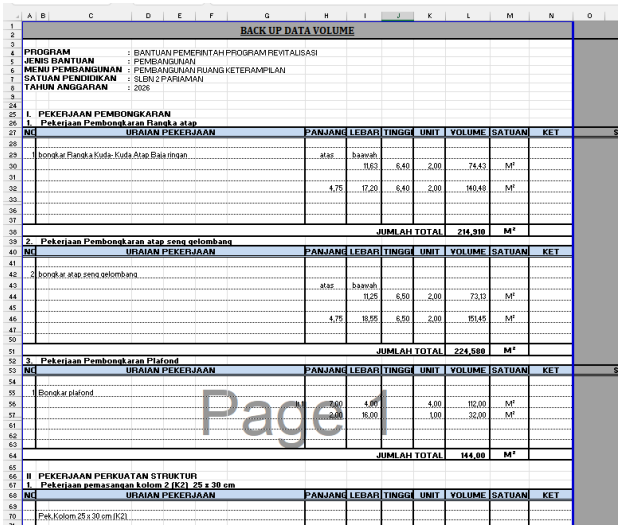
NPSN	10307423
Nama Sekolah	SLBN 2 PARIAMAN
Kabupaten/Kota	PARIAMAN
Provinsi	SUMATERA BARAT
Pekerjaan	PEMBANGUNAN RUANG KETERAMPILAN
Lokasi	JL. A. Y ANI
Tahun Anggaran	2026

No.	URAIAN PEKERJAAN	RAB PERENCANAAN Volume	Sat	Harga Satuan (Rp)	Jumlah Harga (Rp)	PEKERJAAN TAMBAH Volume	Harga Satuan (Rp)	Jumlah Harga (Rp)	PEKERJAAN KURANG Volume	Harga Satuan (Rp)	Jumlah Harga (Rp)	RAB PERUBAHAN Volume	Harga Satuan (Rp)	Jumlah Harga (Rp)	KET
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	PEKERJAAN PERSIAPAN														
1	Pekerjaan pembongkaran plafon dan atap rangka baja	1,00	Ls	3.500.000,00	3.500.000,00				1,00	3.500.000,00	3.500.000,00				Pengurangan Volume
2	Pekerjaan persiapan dan K3		Ls			1,00	4.690.250,00	4.690.250,00			-	1,00	4.690.250,00	4.690.250,00	Penambahan volume
3	Pekerjaan pembongkaran beton dan bata		Ls			1,00	1.500.000,00	1.500.000,00			-	1,00	1.500.000,00	1.500.000,00	Penambahan volume
4	Pekerjaan pembongkaran rangka atap		m2			214,31	14.945,00	3.211.623,95			-	214,31	14.945,00	3.211.623,95	Penambahan volume
5	Pekerjaan pembongkaran plafond		m2			144,00	18.163,00	2.615.472,00			-	144,00	18.163,00	2.615.472,00	Penambahan volume
6	Pekerjaan pembongkaran seng gelombang		m2			224,58	8.261,90	1.855.457,50			-	224,58	8.261,90	1.855.457,50	Penambahan volume
				Jumlah A	3.500.000,00			13.873.009,45						13.873.009,45	
B	PEKERJAAN PERKUATAN STRUKTUR lt 1										-				
1	Pekerjaan Kolom 25 X 30 cm (K2)										-				
	a. Membuat beton Fc 19,3 (K225)		m3			0,45	1.119.797,58	506.719,60			-	0,45	1.119.797,58	506.719,60	Penambahan volume dan penyesuaian ahsp
	b. Pembesian Besi beton ulir		kg			508,86	18.560,56	9.444.664,09			-	508,86	18.560,56	9.444.664,09	Penambahan volume dan penyesuaian ahsp
	c. Pembesian Besi beton Polos		kg			420,72	17.802,31	7.489.863,70			-	420,72	17.802,31	7.489.863,70	Penambahan volume dan penyesuaian ahsp
	d. Bekisting kolom (3x pakai)		m2			81,40	197.556,06	16.081.226,23				81,40	197.556,06	16.081.226,23	Penambahan volume dan penyesuaian ahsp
											-				
2	Pekerjaan Balok (15 x 40 cm)										-				
	a. Membuat beton Fc 19,3 (K225)		m3			1,75	1.119.797,58	1.958.122,85			-	1,75	1.119.797,58	1.958.122,85	Penambahan volume dan penyesuaian ahsp
	b. Pembesian Besi beton ulir		kg			854,97	18.560,56	15.868.737,41			-	854,97	18.560,56	15.868.737,41	Penambahan volume dan penyesuaian ahsp
	c. Pembesian Besi beton Polos		kg			411,94	17.802,31	7.335.156,62			-	411,94	17.802,31	7.335.156,62	Penambahan volume dan penyesuaian ahsp
	d. Bekisting balok (3x pakai)		m2			74,79	197.556,06	14.775.359,68				74,79	197.556,06	14.775.359,68	Penambahan volume dan penyesuaian ahsp
											-				
3	Pekerjaan Balok Anak (15 x 20 cm)										-				
	a. Membuat beton Fc 19,3 (K225)		m3			0,40	1.271.936,25	507.502,56			-	0,40	1.119.797,58	446.799,24	Penambahan volume dan penyesuaian ahsp
	b. Pembesian Besi beton ulir		kg			114,61	18.560,56	2.127.176,60			-	114,61	18.560,56	2.127.176,60	Penambahan volume dan penyesuaian ahsp
	c. Pembesian Besi beton Polos		kg			81,66	17.802,31	1.453.786,69			-	81,66	17.802,31	1.453.786,69	Penambahan volume dan penyesuaian ahsp
	d. Bekisting balok (3x pakai)		m2			12,39	197.556,06	2.447.326,15			-	12,39	197.556,06	2.447.326,15	Penambahan volume dan penyesuaian ahsp

1. Contract Change Order (CCO) Addendum Documentation & Contract Value Variation Summary



2. Revised Architectural Elevation Working Drawings (Front & Rear Elevations)



BACK UP DATA VOLUME

PROGRAM	: BANTUAN PEMERINTAH PROGRAM REVITALISASI
JENIS BANTUAN	: PEMBANGUNAN
MENU PEMBANGUNAN	: PEMBANGUNAN RUANG KETERAMPILAN
SATUAN PENDIDIKAN	: SLBN 2 PARIAMAN
TAHUN ANGGARAN	: 2026

I. PEKERJAAN PEMBONGKARAN

1. Pekerjaan Pembongkaran Rangka atap

NO	URAIAN PEKERJAAN	PANJANG	LEBAR	TINGGI	UNIT	VOLUME	SATUAN	KET
1	Bongkar Rangka Kuda- Kuda Atap Baja ringan	atas	baawah					
			11,63	6,40	2,00	74,43	M²	
		4,75	17,20	6,40	2,00	140,48	M²	
					JUMLAH TOTAL	214,910	M²	

2. Pekerjaan Pembongkaran atap seng gelombang

NO	URAIAN PEKERJAAN	PANJANG	LEBAR	TINGGI	UNIT	VOLUME	SATUAN	KET
1	Bongkar atap seng gelombang	atas	baawah					
			11,25	6,50	2,00	73,13	M²	
		4,75	18,55	6,50	2,00	151,45	M²	
					JUMLAH TOTAL	224,580	M²	

3. Pekerjaan Pembongkaran Plafond

NO	URAIAN PEKERJAAN	PANJANG	LEBAR	TINGGI	UNIT	VOLUME	SATUAN	KET
1	Bongkar plafond							
		7,00	4,00		4,00	112,00	M²	
		2,00	16,00		1,00	32,00	M²	
					JUMLAH TOTAL	144,00	M²	

II PEKERJAAN PERKUATAN STRUKTUR

1. Pekerjaan pemasangan kolom 2 (K2) 25 x 30 cm

NO	URAIAN PEKERJAAN	PANJANG	LEBAR	TINGGI	UNIT	VOLUME	SATUAN	KET
1	Pek.Kolom 25 x 30 cm (K2)							

Page 1

3. Backup Volume Mathematical Takeoff for CCO Addition & Deduction Items

FREELANCE QUANTITY SURVEYOR 03

PT Mulia Djasmine Sentosa

PT Perfetti Van Melle Project

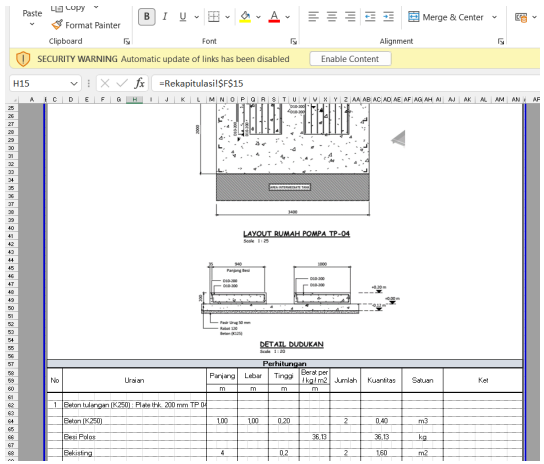
Role:	QS (Freelance)
Contractor:	PT Mulia Djasmine
Client:	PT Perfetti Van Melle
Deliverables:	MC-100, BBS, CCO

QUANTITY SURVEYING SCOPE

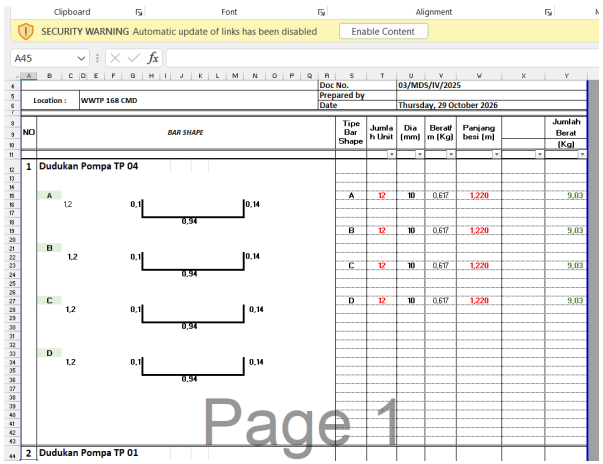
Subcontractor volume recalculation for MC-100, cost variance reconciliation, CCO management, and installed rebar BBS.

PROFESSIONAL FIELD QS ROLE:

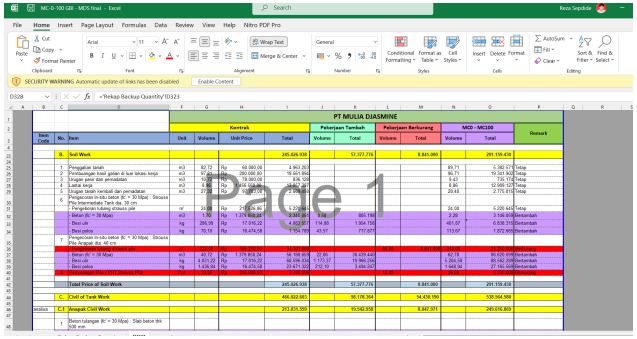
- Controlling subcontractor cash flow through joint on-site physical progress surveys with the main contractor.
- Ensuring progress payment billings align strictly with installed physical quantities without unjustified discrepancies.



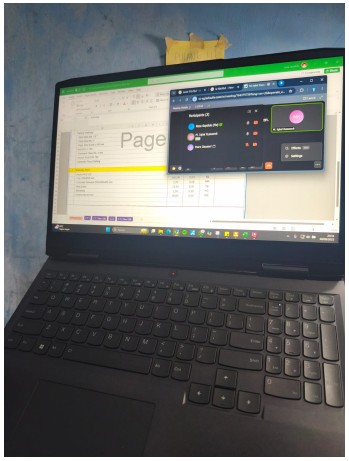
1. MC-100 Final Quantity Take-Off & Joint Measurement Survey Backup



2. Bar Bending Schedule (BBS) Rebar Cutting Optimization



3. Contract Change Order (CCO) Item Variation Summary



4. Coordination Meeting & Joint Site Measurement Inspection

D3 CIVIL ENGINEERING FINAL THESIS 04

DPRD Pasaman Building

District of Pasaman, West Sumatra

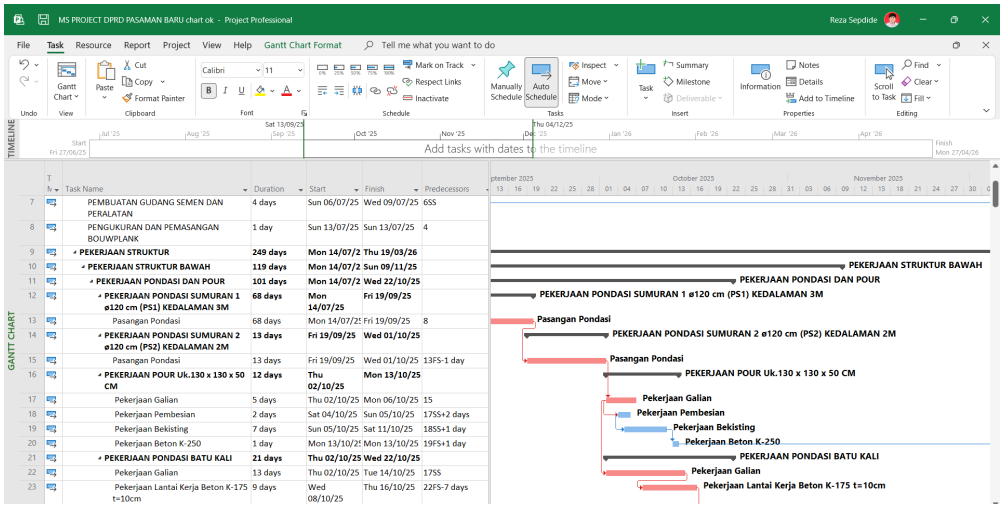
Institution:	Padang State Polytechnic
Study Object:	DPRD Pasaman Building
Total Budget:	Rp 16,042,000,000,00
Duration:	240 Calendar Days
Software:	MS Project & MS Excel

PLANNING SCOPE

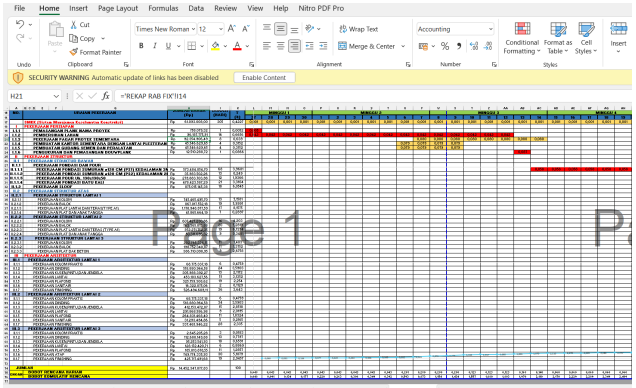
- **Work Breakdown Structure (WBS):** Hierarchical decomposition of caisson deep foundations, columns, beams, floor slabs, and structural steel roof framing.
- **Cost Estimation (BoQ/RAB):** Application of PUPR Unit Price Analysis (AHSP) integrated with regional labor, material, and heavy equipment market surveys.
- **S-Curve Progress Baseline:** Simulation of weekly cumulative financial weight distribution across a 240-calendar-day schedule.

SCHEDULE & COST INTEGRATION:

- **Critical Path Method (CPM):** Precedence network logic scheduling (*Finish-to-Start*, lag times) identifying critical path activities across 34-week execution.
- **Progress & Cost Control:** S-Curve baseline serving as executive instrument to track physical milestones and eliminate cost variances.
- **Budget Reliability:** Total cost estimate of Rp 16.04 Billion rigorously verified through comprehensive structural & architectural BoQ summaries.



1. Work Breakdown Structure (WBS) & Microsoft Project Gantt Chart Baseline (240 Calendar Days)



2. S-Curve Cumulative Weekly Physical Weight Distribution Baseline

Tabel 3.19 Rekapitulasi Rencana Anggaran Biaya

REKAPITULASI RENCANA ANGGARAN BIAYA

Proyek : Rencana Pelaksanaan Pembangunan Gedung DPRD Kabupaten Pasaman
Lokasi : Jl.Jend.Sudirman No.29,Pauah,Kec.Lubuk Sikaping,Kabupaten Pasaman,Sumatera Barat 26318

NO.	URAIAN PEKERJAAN	SUB TOTAL	
	SMKK (Sistem Manajemen Keselamatan Konstruksi)	Rp	61.093.008,00
I	PEKERJAAN PERSIAPAN	Rp	292.904.309,73
II	PEKERJAAN STRUKTUR		
II.1	PEKERJAAN STRUKTUR BAWAH		
II.1.1	PEKERJAAN PONDASI DAN POUR	Rp	1.367.775.648,24
II.1.2	PEKERJAAN SLOOF	Rp	875.015.145,38
II.2	PEKERJAAN STRUKTUR ATAS		
II.2.1	PEKERJAAN STRUKTUR LANTAI 1	Rp	2.833.297.689,66
II.2.2	PEKERJAAN STRUKTUR LANTAI 2	Rp	2.393.375.693,41
II.2.3	PEKERJAAN STRUKTUR LANTAI 3	Rp	788.227.773,97
III	PEKERJAAN ARSITEKTUR		
III.1	PEKERJAAN ARSITEKTUR LANTAI 1	Rp	2.215.324.314,38
III.2	PEKERJAAN ARSITEKTUR LANTAI 2	Rp	1.925.269.271,09
III.3	PEKERJAAN ARSITEKTUR LANTAI 3	Rp	1.700.065.023,96
	Jumlah Sub Total	Rp	14.452.347.877,83
	PPN 11%	Rp	1.589.758.266,56
	Total Biaya	Rp	16.042.106.144,39
	Dibulatkan	Rp	16.042.000.000,00

TERBILANG : ENAM BELAS MILIYAR LIMA RATUS DELAPAN PULUH DUA JUTA RUPIAH

3. Detailed Bill of Quantities (BoQ) & Cost Summary (Total: Rp 16.04 Billion)

INDUSTRIAL INTERNSHIP PROJECT 05

Waste Gas Recovery Project

PT Indo-Bharat Rayon Purwakarta

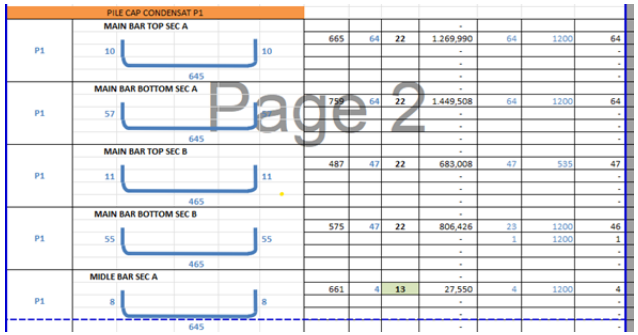
Contractor:	PT Solusi Industri Energi
Main Client:	PT Indo-Bharat Rayon
Project Location:	Purwakarta, West Java
Activity Focus:	Rebar BBS & Site Logistics

BBS DRAFTING & FIELD SUPERVISION

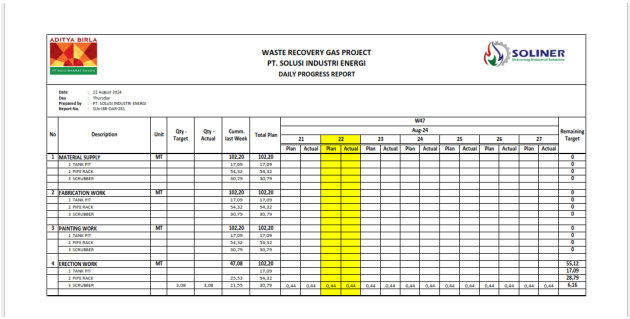
- **Bar Bending Schedule (BBS):** Detailed rebar cutting and bending schedules for WGR heavy machine foundations.
- **Scrap Waste Optimization:** Optimization of stock bar cut patterns keeping rebar scrap loss below $< 3\%$.
- **Daily Progress Logs:** Logging daily fabrication output, rebar assembly metrics, and on-site field verification.

INDUSTRIAL REBAR STANDARDS:

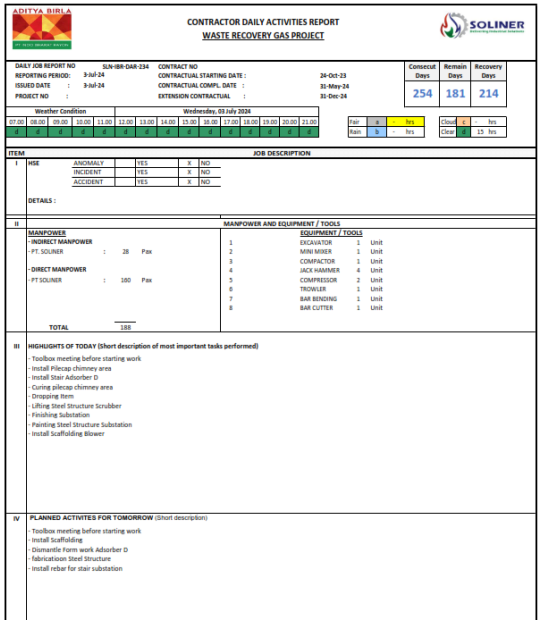
- **Bending Precision:** Heavy vibrating machine foundation assembly demanding $135°$ seismic hook geometries and SNI 2847 development lengths.
- **Cage Rigidity:** Double-strand tie-wire tying ensuring rigid rebar cages without displacement during heavy concrete pouring.
- **Site HSE Rigor:** Daily supervision adhering strictly to rigorous industrial plant chemical HSE safety regulations.



1. Bar Bending Schedule (BBS) Rebar Cutting Optimization



2. Daily Rebar Fabrication Progress Tracking Sheet



3. Field Daily Inspection & Fabrication Work Log

4. On-Site Field Team Coordination & Documentation Management

Field coordination of PT Solusi Industri Energi engineering team on WGR plant site

REWORK CALCULATION & REPORTING 05

QTO Rework & Reporting

Field Operation Control

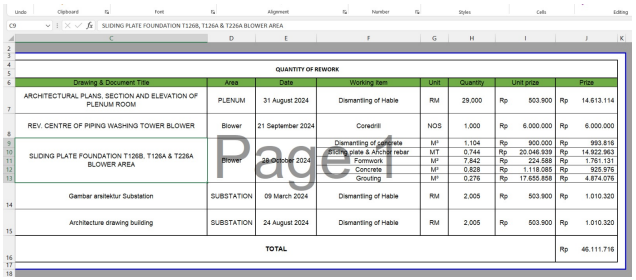
Activity 1:	Volume & Rework Calculation
Activity 2:	Progress Billing Backup
Reporting:	Daily, Weekly, Monthly
Technical Impact:	Accurate Cost Reconciliation

FIELD CONTROL RESPONSIBILITY

- **Rework Quantity Take-Off:** Mathematical recalculation of structural modifications and machine foundation rectifications.
- **Progress Billing Backup:** Formulating detailed quantity takeoff sheets supporting monthly client progress payment certificates.
- **Joint Site Measurement:** Periodic physical site verification and joint measurement surveys with Owner and Construction Management consultants.

REPORTING ACCOUNTABILITY:

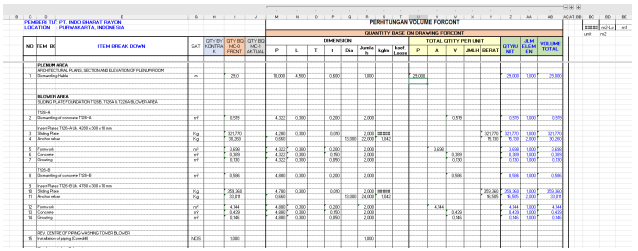
- **Hierarchical Reporting:** Daily logs systematically compiled into weekly and monthly reports as legal progress records.
- **Variation Reconciliation:** Orderly backup documentation preventing claim disputes between subcontractors and the main contractor.
- **Manpower Tracking:** Daily workforce allocation tracking (peak 254 personnel) for productivity control and HSE compliance.



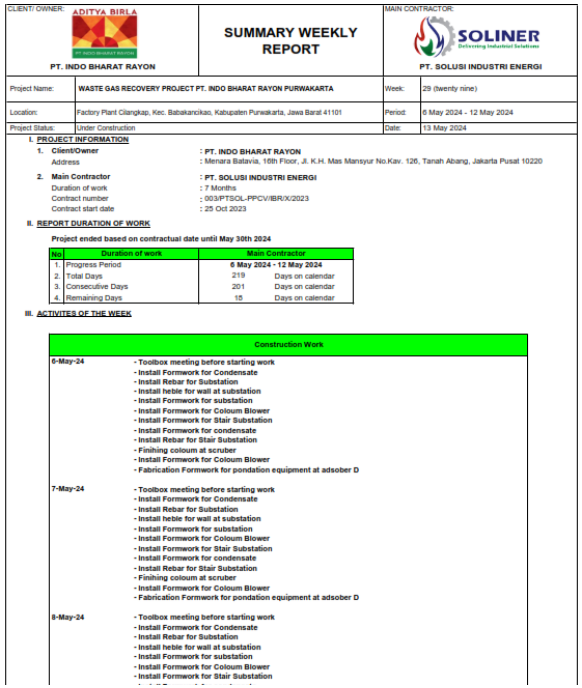
QUANTITY OF REWORK

Drawing & Document Title	Area	Date	Working Item	Unit	Quantity	Unit prize	Prize
ARCHITECTURAL PLANS, SECTION AND ELEVATION OF PLENUM ROOM	PLENUM	31 August 2024	Dismantling of Hable	RM	29,000	Rp 503.900	Rp 14.613.114
REV. CENTRE OF PIPING WASHING TOWER BLOWER	Blower	21 September 2024	Coredrill	NOS	1,000	Rp 6.000.000	Rp 6.000.000
SLIDING PLATE FOUNDATION T126B, T126A & T226A BLOWER AREA	Blower	28 October 2024	Dismantling of concrete Sliding plate & Achor rebar Formwork Concrete Grouting	M³ MT M² M³ M³	1,104 0,744 7,842 0,828 0,276	Rp 900.000 Rp 20.046.939 Rp 224.588 Rp 1.118.085 Rp 17.655.858	Rp 993.816 Rp 14.922.963 Rp 1.761.131 Rp 925.976 Rp 4.874.076
Gambar arsitektur Substation	SUBSTATION	09 March 2024	Dismantling of Hable	RM	2,005	Rp 503.900	Rp 1.010.320
Architecture drawing building	SUBSTATION	24 August 2024	Dismantling of Hable	RM	2,005	Rp 503.900	Rp 1.010.320
TOTAL							Rp 46.111.716

1. Rework Bill of Quantities (BoQ) & Cost Take-Off



2. Rework Quantity Take-Off (QTO) Backup Volume Sheet



CLIENT/ OWNER: ADITYA BIRLA — PT. INDO BHARAT RAYON

SUMMARY WEEKLY REPORT

MAIN CONTRACTOR: SOLINER — PT. SOLUSI INDUSTRI ENERGI

Project Name:	WASTE GAS RECOVERY PROJECT PT. INDO BHARAT RAYON PURWAKARTA	Week:	29 (twenty nine)
Location:	Factory Plant Cilangkap, Kec. Babakancikao, Kabupaten Purwakarta, Jawa Barat 41101	Period:	6 May 2024 - 12 May 2024
Project Status:	Under Construction	Date:	13 May 2024

I. PROJECT INFORMATION

1. Client/Owner : PT. INDO BHARAT RAYON
   Address : Menara Batavia, 16th Floor, Jl. K.H. Mas Mansyur No.Kav. 126, Tanah Abang, Jakarta Pusat 10220
2. Main Contractor : PT. SOLUSI INDUSTRI ENERGI
   Duration of work : 7 Months
   Contract number : 003/PTSOL-PPCV/IBR/X/2023
   Contract start date : 25 Oct 2023

II. REPORT DURATION OF WORK

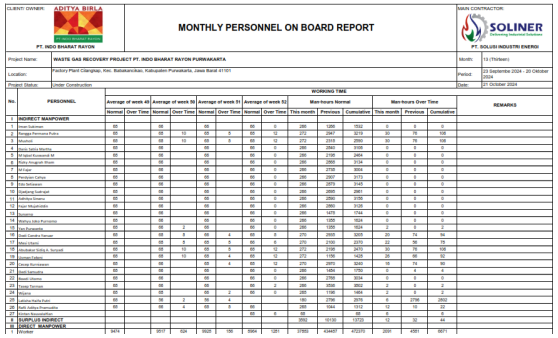
Project ended based on contractual date until May 30th 2024

No	Duration of work	Main Contractor
1.	Progress Period	6 May 2024 - 12 May 2024
2.	Total Days	219 Days on calendar
3.	Consecutive Days	201 Days on calendar
4.	Remaining Days	15 Days on calendar

III. ACTIVITES OF THE WEEK

Construction Work	
6-May-24	- Toolbox meeting before starting work - Install Formwork for Condensate - Install Rebar for Substation - Install hebie for wall at substation - Install Formwork for substation - Install Formwork for Coloum Blower - Install Formwork for Stair Substation - Install Formwork for condensate - Install Rebar for Stair Substation - Finihing coloum at scruber - Install Formwork for Coloum Blower - Fabrication Formwork for pondation equipment at adsober D
7-May-24	- Toolbox meeting before starting work - Install Formwork for Condensate - Install Rebar for Substation - Install hebie for wall at substation - Install Formwork for substation - Install Formwork for Coloum Blower - Install Formwork for Stair Substation - Install Formwork for condensate - Install Rebar for Stair Substation - Finihing coloum at scruber - Install Formwork for Coloum Blower - Fabrication Formwork for pondation equipment at adsober D
8-May-24	- Toolbox meeting before starting work - Install Formwork for Condensate - Install Rebar for Substation - Install hebie for wall at substation - Install Formwork for substation - Install Formwork for Coloum Blower - Install Formwork for Stair Substation

3. Field Weekly Construction Progress Report



CLIENT/ OWNER: ADITYA BIRLA — PT. INDO BHARAT RAYON

MONTHLY PERSONNEL ON BOARD REPORT

MAIN CONTRACTOR: SOLINER — PT. SOLUSI INDUSTRI ENERGI

Project Name:	WASTE GAS RECOVERY PROJECT PT. INDO BHARAT RAYON PURWAKARTA
Location:	Factory Plant Cilangkap, Kec. Babakancikao, Kabupaten Purwakarta, Jawa Barat 41101
Project Status:	Under Construction
Month:	13 (Thirteen)
Period:	23 September 2024 - 20 October 2024
Date:	21 October 2024

No.	PERSONNEL	WORKING TIME: Average of week 49 Normal	Average of week 49 Over Time	Average of week 50 Normal	Average of week 50 Over Time	Average of week 51 Normal	Average of week 51 Over Time	Average of week 52 Normal	Average of week 52 Over Time	Man-hours Normal: This month	Man-hours Normal: Previous	Man-hours Normal: Cumulative	Man-hours Over Time: This month	Man-hours Over Time: Previous	Man-hours Over Time: Cumulative	REMARKS
1	INDIRECT MANPOWER															
2	Irwan Gultomm	65	0	65	0	65	0	65	0	260	1260	1552	0	0	0	
3	Hengga Permana Putra	65	0	65	10	65	5	65	12	272	2047	2319	30	78	108	
4	Mochdi	65	0	65	10	65	5	65	12	272	2319	2591	30	78	108	
5	Dewi Ratna Martha	65	0	65	0	65	0	65	0	260	2864	3124	0	0	0	
6	Ari Ignat Kurnianto M	65	0	65	0	65	0	65	0	260	2125	2464	0	0	0	
7	Rino Anugrah Risan	65	0	65	0	65	0	65	0	260	2864	3124	0	0	0	
8	Ari Igar	65	0	65	0	65	0	65	0	260	2730	3004	0	0	0	
9	Rangga Cahya	65	0	65	0	65	0	65	0	260	2597	3173	0	0	0	
10	Dio Setiawan	65	0	65	0	65	0	65	0	260	2873	3145	0	0	0	
11	Dwiyang Sukriyat	65	0	65	0	65	0	65	0	260	2600	2861	0	0	0	
12	Ivanhoe Gracia	65	0	65	0	65	0	65	0	260	2860	3120	0	0	0	
13	Irsan Mulyadinata	65	0	65	0	65	0	65	0	260	2860	3120	0	0	0	
14	Sandung	66	0	66	0	66	0	66	0	266	1478	1744	0	0	0	
15	Aditya Abu Rahmadia	65	0	65	0	65	0	65	0	260	1300	1624	0	0	0	
16	Dwi Kurniadi	65	0	65	2	65	0	65	0	260	1300	1624	2	0	2	
17	Saad Zamhir Damas	65	0	65	8	65	4	65	6	270	2555	3205	20	74	94	
18	Wisnu Utama	65	0	65	8	65	5	65	6	270	2105	2370	22	56	78	
19	Muhamad Gilang N. Sumyadi	65	0	65	10	65	5	65	12	272	2138	2470	30	78	108	
20	Hendra Cahaya	65	0	65	10	65	4	65	12	270	1555	1825	26	66	92	
21	Dessa Rumiasan	65	0	65	8	65	4	65	12	270	2970	3240	16	74	90	
22	Saad Samudra	66	0	66		66		66	0	266	1494	1760	0	4	4	
23	Saad Usama	66	0	66		66		66	0	266	2794	3054	0	0	0	
24	Tessa Ramsan	66	0	66		66		66	0	266	2658	3052	2	0	2	
25	Wisnu	65	0	65		65	2	65	0	260	1196	1456	2	0	2	
26	Juliana Nella Putri	65	0	65	2	65	4			195	2784	2979	6	2784	2850	
27	Bahi Aditya Pramudita	65	0	65	4	65	0	65		260	1044	1312	12	10	22	
28	Ahmad Fauzi Ramadhan							65	0	65		65	0	0	0	
29	SUBTOTAL INDIRECT									3952	50130	53731	72	32	44	
30	DIRECT MANPOWER															
31	Konsel	8474	0	8517	324	8525	156	8484	1281	31950	424451	456201	2331	4581	6911	

4. Executive Monthly Construction Progress Report

Monthly Manpower & Progress Recapitulation:

- Peak manpower: 254 coordinated site personnel.
- Monthly verification supporting progress billing certificates.
- Field safety performance achieving Zero LTI status.

FIELD QUALITY ASSURANCE & CONTROL

QC Inspection & Handover Protocols

PT Indo-Bharat Rayon Plant Site

Concrete Testing:	Fresh Concrete Slump Test
Rebar Inspection:	Pre-Pour Verification Checklist
Zoning:	Field Work Area Mapping
Handover:	Punch List Defect Rectification

PRACTICAL QC FIELD ACTIVITIES

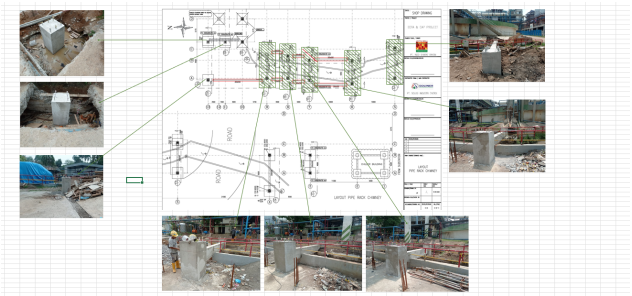
Fresh concrete slump testing per mixer truck batch, rebar placement verification, concrete cover clearance, pre-pour formwork cleanliness, work area zoning, and punch list rectification tracking.

QUALITY CONTROL INTEGRITY PRINCIPLES:

- Strict zero-tolerance: No concrete pouring permitted before slump test passes specification (12 ± 2 cm).
- Pre-pour rebar checklist and formwork cleanliness sign-off mandatory prior to batch release.

1. Fresh Concrete Slump Cone Workability Testing

2. Handover Punch List & Defect Rectification Log



3. Heavy Industrial Work Area Mapping & Zoning Plan

4. Field Quality Checklist Inspection & Documentation

GOVERNMENT GRANT FACILITATOR 06

Field Technical Facilitator

Kemendikdasmen RI Program

Agency:	Ministry of Primary and Secondary Education (Kemendikdasmen)
Assignment:	Technical School Facilitator
Period:	June — December 2025
Scope:	Beneficiary Schools

FACILITATION RESPONSIBILITIES

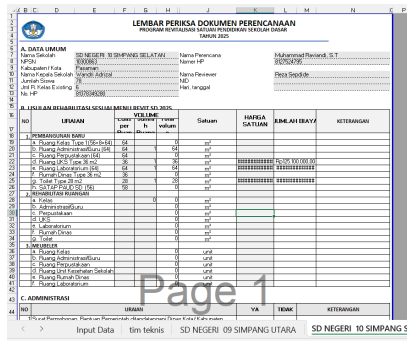
Feasibility reviews of DED drawings, BoQ/RAB verification for beneficiary schools, monitoring physical milestones, disbursement verification, and technical bimtek guidance.

PROGRAM STRATEGIC OVERSIGHT:

- Ensuring state grant funding meets designated physical targets, engineering quality standards, and strict timelines.
- Enforcing formal Milestone Acceptance Minutes (BA) for national state financial audit compliance and accountability.

1. Initial Field Structural Monitoring Inspection

2. Ministry Technical Briefing & Facilitation Workshop



3. Technical Document Review & Verification Form

A. DAFTAR SEKOLAH YANG DIDAMPINGI

No	Nama Sekolah	NPSN	Kab/Kota	Provinsi
1	SD NEGERI 09 SIMPANG UTARA	10300742	Pasaman	Sumatera Barat
2	SD NEGERI 19 PARIT LUBANG	10300784	Pasaman	Sumatera Barat
3	SD NEGERI 05 SERASI	10300815	Pasaman	Sumatera Barat
4	SD NEGERI 10 SIMPANG SELATAN	10300863	Pasaman	Sumatera Barat
5	SD NEGERI 02 SIMPANG TIGA CUBADAK	10300904	Pasaman	Sumatera Barat
6	SD NEGERI 04 SETIA	10300931	Pasaman	Sumatera Barat

B. KEGIATAN FASILITATOR

1. PRA PELAKSANAAN PROGRAM REVITALISASI

a. TOT Fasilitator dan Koordinator

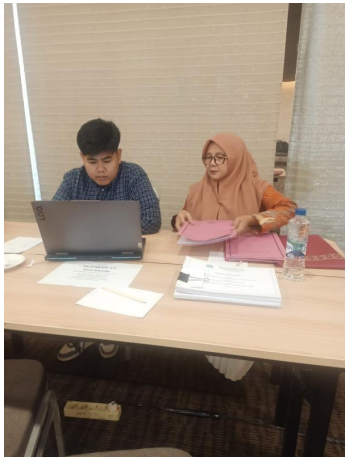
1) Deskripsi Keikutsertaan TOT
Kegiatan Training of Trainer yang dilakukan pada tanggal 16 s.d 19 Juni 2025 di Grand Mercure Harmoni Hayam Wuruk, Gambir, Jakarta. TOT adalah kegiatan pelatihan yang bertujuan untuk memandu, mengarahkan dan meningkatkan kapasitas peserta dalam memahami Langkah kerja dan tujuan pelaksanaan program revitalisasi

2) Hasil Kegiatan TOT
Mengetahui apa yang akan menjadi tugas sebagai fasilitator dan arahan pekerjaan yang akan dilakukan saat dimulai revitalisasi, meningkatkan kompetensi peserta dalam bidang manajemen,pembelajaran berbasis proyek, pendidikan karakter dan oeningkatan SD.

b. Bimtek Sekolah Calon Penerima Bantuan

1) Deskripsi Keikutsertaan Bimtek (daring dan luring)
Kegiatan Training of Trainer yang dilakukan pada tanggal 16 s.d 19 Juni 2025 di Grand Mercure Harmoni Hayam Wuruk, Gambir, Jakarta. TOT adalah kegiatan pelatihan yang bertujuan untuk memandu, mengarahkan dan meningkatkan kapasitas peserta dalam memahami Langkah kerja dan tujuan pelaksanaan

4. Facilitator Monthly Physical Progress Report



5. Technical Reviewer Stakeholder Coordination Meeting

6. Comprehensive Site Inspection & Milestone Verification

REVIT BIM PARAMETRIC MODELING

Digital 3D BIM Modeling

Revit Masterclass Academy

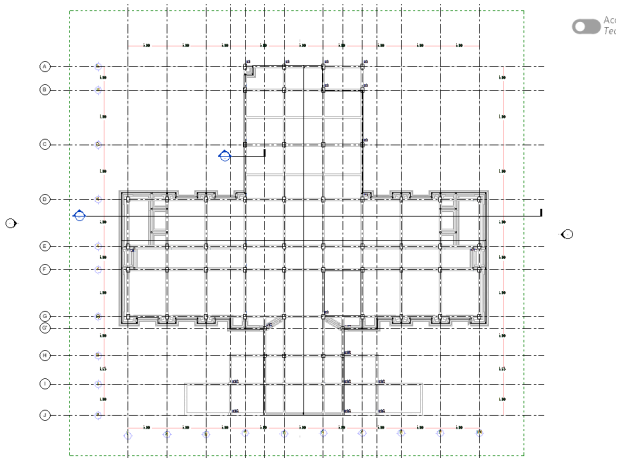
Platform:	Autodesk Revit
Pelatihan:	Revit Masterclass
Academy:	Arsisten Academy
Empirical Site:	Struktur & Arsitektur

METHODOLOGY & MODELING DELIVERABLES

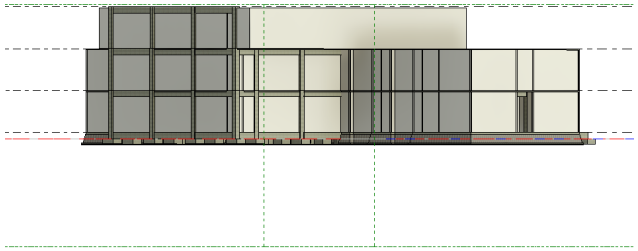
3D parametric modeling of structural components (columns, beams, slabs) and architectural elements, re-modeling from 2D DED, coordinated view templates, and automated material take-off schedules.

BIM 3D DIGITALIZATION PRINCIPLES:

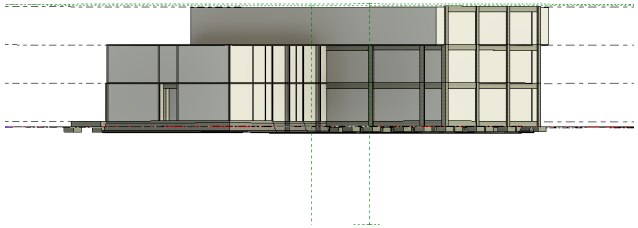
- Deep comprehension of parametric 3D data integration with construction scheduling and quantity takeoff.
- Capable of generating coordinated floor plans, elevations, and section cuts automatically with zero graphical mismatch.



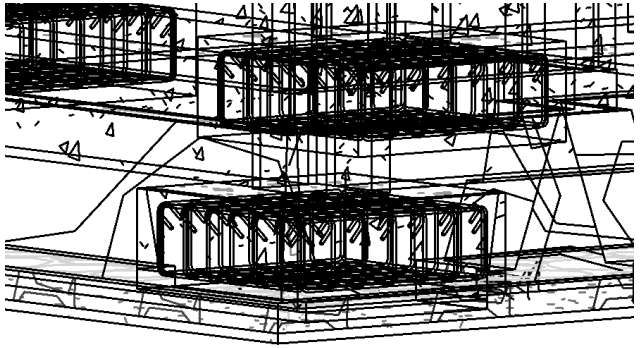
1. 3D Isometric View: Structural Reinforced Concrete Frame



2. Front Architectural Facade & Glazing Openings



3. Transverse Section Cut & Beam-Column Joint Details



4. Coordinated Architectural & Structural Isometric Perspective

STRUCTURAL ENGINEERING ANALYSIS 08

SAP2000 3D Finite Element Analysis

Multi-Story Concrete Building Frame

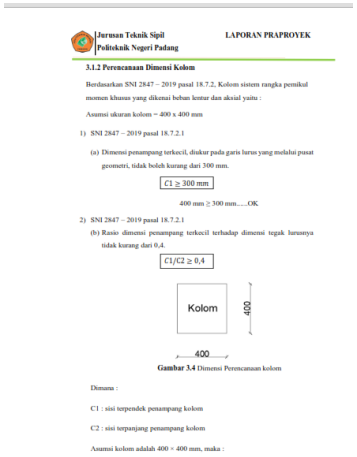
Platform:	CSI SAP2000
Scheme:	Multi-Story Building Design Pre-Project
Institution:	Padang State Polytechnic
Analysis Type:	Internal Forces & SNI Dynamic Seismic Loading

STRUCTURAL ENGINEERING SCOPE

3D reinforced concrete space frame modeling, dead load (DL), live load (LL), SNI 1726 response spectrum seismic loading, bending moment (M_3), shear force (V_2), and axial force (P) diagrams.

SEISMIC DESIGN CODE COMPLIANCE:

- Earthquake-resistant capacity evaluation for reinforced concrete structures conforming to SNI 1726:2019.
- Application of load combinations (1.2D + 1.6L + 1.0E) under Special Moment Resisting Frame Systems (SRPMK).



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LAPORAN PRAPROYEK

3.1.2 Perencanaan Dimensi Kolom

Berdasarkan SNI 2847 – 2019 pasal 18.7.2, Kolom sistem rangka pemikul momen khusus yang dikenai beban lentur dan aksial yaitu :

Asumsi ukuran kolom = 400 x 400 mm

1) SNI 2847 – 2019 pasal 18.7.2.1

(a) Dimensi penampang terkecil, diukur pada garis lurus yang melalui pusat geometri, tidak boleh kurang dari 300 mm.

C1 ≥ 300 mm

400 mm ≥ 300 mm.....OK

2) SNI 2847 – 2019 pasal 18.7.2.1

(b) Rasio dimensi penampang terkecil terhadap dimensi tegak lurusnya tidak kurang dari 0,4.

C1/C2 ≥ 0,4

Gambar 3.4 Dimensi Perencanaan kolom

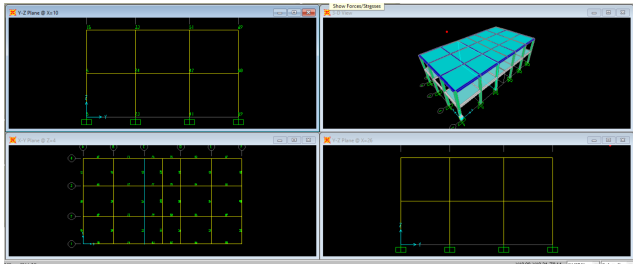
Dimana :

C1 : sisi terpendek penampang kolom

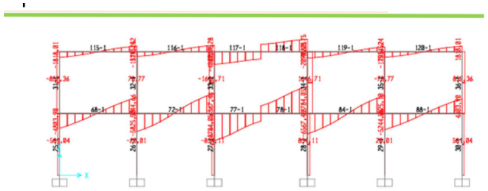
C2 : sisi terpanjang penampang kolom

Asumsi kolom adalah 400 × 400 mm, maka :

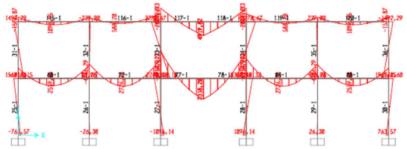
1. 3D Concrete Space Frame Structural Geometry



2. SAP2000 Finite Element Modeling Interface

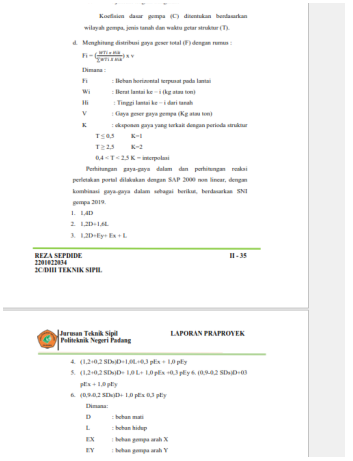


Gambar 3. 76 Gaya Geser Kombinasi 3



Gambar 3. 77 Momen Kombinasi 3

3. Diagram Momen & Gaya Geser (BMD/SFD)



Koefisien dasar gempa (C) ditentukan berdasarkan wilayah gempa, jenis tanah dan waktu getar struktur (T).

d. Menghitung distribusi gaya geser total (F) dengan rumus :

Dimana :

Fi : Beban horizontal terpusat pada lantai

Wi : Berat lantai ke - i (kg atau ton)

Hi : Tinggi lantai ke - i dari tanah

V : Gaya geser gaya gempa (Kg atau ton)

K : eksponen gaya yang terkait dengan periode struktur

T ≤ 0,5 K=1

T ≥ 2,5 K=2

0,4 < T < 2,5 K = interpolasi

Perhitungan gaya-gaya dalam dan perbitungan reaksi perletakan portal dilakukan dengan SAP 2000 non linear, dengan kombinasi gaya-gaya dalam sebagai berikut, berdasarkan SNI gempa 2019.

1. 1,4D
2. 1,2D+1,6L
3. 1,2D+Ey+ Ex + L

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LAPORAN PRAPROYEK

4. (1,2+0,2 SDs)D+1,0L+0,3 ρEx + 1,0 ρEy
5. (1,2+0,2 SDs)D+ 1,0 L+ 1,0 ρEx +0,3 ρEy 6. (0,9-0,2 SDs)D+03 ρEx + 1,0 ρEy
6. (0,9-0,2 SDs)D- 1,0 ρEx 0,3 ρEy

Dimana:

D : beban mati

L : beban hidup

EX : beban gempa arah X

EY : beban gempa arah Y

4. Input Parameter Beban Gempa SNI

NATIONAL COMPETITION 09

Fondasi Days 2025

Universitas Sebelas Maret (UNS)

Implementing Agency:	Univ. Sebelas Maret
Delegation Team:	Jaya Jaya Team (PNP)
Competition Category:	Cost Estimation & Tender Bidding
Deliverables:	Technical Tender Bid, BoQ, S-Curve

TEAM DELIVERABLES & ENGINEERING RIGOR

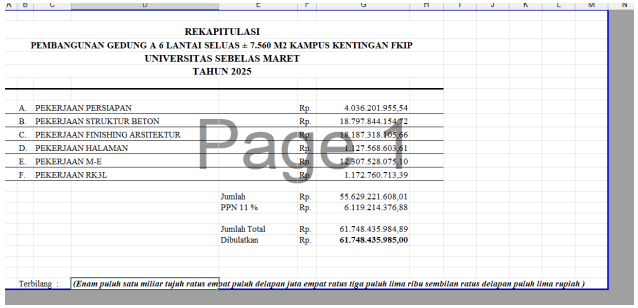
Technical tender bid for 4-story FKIP UNS building, competitive quantity takeoff & BoQ, 3D SketchUp construction execution staging, CPM S-Curve, and executive bidding presentation.

TENDER BIDDING STRATEGY:

- Structuring competitive bid pricing via lean construction methods and rigorous procurement administrative compliance.
- Critical Path Method (CPM) optimization to minimize contractor indirect overhead expenses.

1. Technical Tender Competition Poster: Jaya-Jaya Team

2. 3D Construction Execution Sequencing & Methodology Model



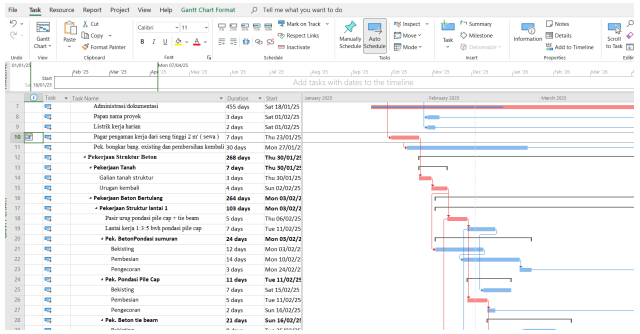
REKAPITULASI
PEMBANGUNAN GEDUNG A 6 LANTAI SELUAS ± 7.560 M2 KAMPUS KENTINGAN FKIP
UNIVERSITAS SEBELAS MARET
TAHUN 2025

A.	PEKERJAAN PERSIAPAN	Rp.	4.036.201.955,54
B.	PEKERJAAN STRUKTUR BETON	Rp.	18.797.844.154,72
C.	PEKERJAAN FINISHING ARSITEKTUR	Rp.	18.187.318.805,66
D.	PEKERJAAN HALAMAN	Rp.	1.127.568.603,61
E.	PEKERJAAN M-E	Rp.	12.507.528.075,10
F.	PEKERJAAN RK3L	Rp.	1.172.760.713,39
	Jumlah	Rp.	55.629.221.608,01
	PPN 11 %	Rp.	6.119.214.376,88
	Jumlah Total	Rp.	61.748.435.984,89
	Dibulatkan	Rp.	61.748.435.985,00

Terbilang : *(Enam puluh satu miliar tujuh ratus empat puluh delapan juta empat ratus tiga puluh lima ribu sembilan ratus delapan puluh lima rupiah)*

Page

3. Competitive Bill of Quantities (BoQ) & Unit Price Analysis



4. CPM S-Curve Scheduling & Microsoft Project Gantt Baseline

SCIENTIFIC PUBLICATION 10

Scaffolding Cost Analysis

PKM-AI Kemendikbudristek

Scheme:	PKM-AI Scientific Article
Focus:	Scaffolding Economics
Empirical Site:	PT Solusi Industri Energi
Methodology:	Capex vs Opex Comparison

SCIENTIFIC RESEARCH FOCUS

Comparative economic and safety analysis between rental versus purchasing schemes and modular versus tubular scaffolding systems on industrial chemical jobsites.

ENGINEERING ECONOMIC FINDINGS:

- Break-Even Point (BEP) achieved at critical utilization duration threshold of ≈ 1.75 months.
- For project durations exceeding 2 months, direct purchasing scheme achieves significant cost savings compared to continuous leasing.

ANALISA BIAYA SCAFFOLDING PADA PROYEK *WASTE GAS RECOVERY* IBR PURWAKARTA

Reza Sepdide[1)] , Yoga Eka Putra[2)] , Humaira Azizah. A[3)]
[1),2),3)]Program Studi DIII Teknik Sipil, Jurusan Teknik Sipil, Politeknik Negeri Padang, Kampus Limau Manis Kota Padang, Sumatera Barat, Indonesia

Penulis korespondensi: enitasuardi@yahoo.com

ABSTRAK

Proyek *Waste Gas Recovery* IBR Purwakarta merupakan proyek swasta di Purwakarta yang membangun konstruksi sipil dengan nilai biaya RP 44.000.000.000, menggunakan scaffolding sebagai salah satu elemen penting dalam konstruksi yang digunakan untuk mendukung pekerjaan di ketinggian. Analisis biaya scaffolding mencakup berbagai komponen, seperti biaya material tenaga kerja, dan peralatan,yang bertujuan untuk memberikan gambaran yang jelas mengenai alokasi biaya dan efisiensi biaya untuk penggunaan scaffolding. Untuk mengetahui efisiensi biaya penggunaan scaffolding secara penyewaan dari bulan januari hingga oktober dengan harga Rp 1.194.173.631 dibandingkan pembelian secara langsung sesuai barang yang disewakan dengan harga Rp 682.719.175. Perbedaan harga terlihat setelah analisa biaya dilakukan dan didapatkan efisiensi biaya pada proyek *Waste Gas Recovery* IBR Purwakarta.

Kata-kata Kunci : Scaffolding, Biaya, Proyek

Scientific Article Manuscript: Economic Analysis of Scaffolding Systems on Waste Gas Recovery Project (PKM-AI)

ENGINEERING ECONOMIC COMPARISON RESULTS (IBR PROJECT CASE STUDY)

Purchasing Scheme (Capex):	Total capital investment Rp 1,194,173,631 (Full asset ownership, residual salvage value retained).
Rental Scheme (Opex):	Operational rental cost Rp 682,719,175 per operational cycle with zero equity recovery.
Break-Even Threshold:	**Critical Duration: ≈ 1.75 Months of Utilization**
Strategic Conclusion:	For project durations exceeding 2 months, purchasing proves **significantly more cost-effective, saving hundreds of millions of rupiah** over recurrent rentals.

DATA AUTHENTICITY STATEMENT

Formal Declaration

Engineering Integrity & Accountability

I hereby declare under professional oath that all qualification records, educational transcripts, BNSP certification credentials, AutoCAD DED sheets, RKS specifications, BoQ/RAB cost models, S-Curve schedules, and field documentation presented herein are **authentic, legally valid, and directly verifiable**.

Padang, West Sumatra, Indonesia, 2026

REZA SEPDIDE, A.Md.T.

Intermediate Building Field Supervisor (BNSP Level 5)

DIRECT COMMUNICATION CHANNELS

Official Contact Details

Direct Inquiry Channels

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+62 896-4459-0915

LINKEDIN:

linkedin.com/in/rezasepdide

RESIDENCE:

Padang, West Sumatra, Indonesia

QUALIFICATION SUMMARY

Engineering Specialization

Quantity Surveyor & Field Supervisor

1. Quantity Surveying & Estimation

Precision QTO, BoQ, PUPR AHSP unit rates, rebar BBS detailing, and CCO addendum cost variance control.

2. Site Supervision & K3 Safety

BNSP Level 5 Intermediate Building Field Supervisor License, SMKK Safety & Concrete Quality Control.

3. Digital BIM & CAD Modeling

Autodesk Revit Masterclass, multi-viewport AutoCAD DED paper space layouts, and SAP2000 3D seismic analysis.

4. Certified Bilingual Communication

Japanese (SSW JAC, JFT-Basic 250, JLPT N4 165) and Professional Working English (TOEIC Prediction 640).

PROFESSIONAL COMMITMENT:

- All supporting diplomas, academic transcripts, BNSP licenses, and language credentials are ready for physical verification.
- Committed to the highest standards of the engineering professional code of ethics and construction data integrity.

MOBILIZATION READINESS:

- **Work Placement:** Available for immediate full-time employment across Indonesia or overseas assignment (Japan / Asia).
- **Field Flexibility:** Ready for on-site jobsite deployment or central technical planning office roles.
- **Dossier Package:** Formal legal credentials, BNSP Level 5 SKK, DED CAD Portfolio, D3 Diploma, and certified language scores ready for dispatch.